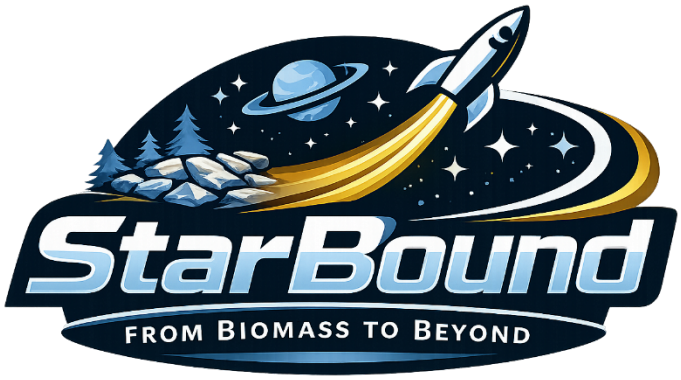


StarBound: From Biomass to Beyond

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Executive summary

The objective of this project is to evaluate the technical feasibility of producing rocket propellant fuel from biomass, specifically woodchips, using a simulated process. Interest in biomass-derived propellants is driven by increasing demand for sustainable aerospace fuels, the availability of feedstocks, and favorable state incentives to support renewable fuel production. This study is intended to support early-stage process design and identify key performance drivers prior to experimental validation or detailed economic analysis.

The proposed process converts woodchip biomass into usable rocket propellant through a sequence of fast pyrolysis, bio-oil upgrading (hydrotreating), and fractionation. Process modeling is currently being conducted using ASPEN simulation software to represent major unit operations, mass and energy balance, cost analysis, utilities, and overall system behavior. At this stage, the simulation focuses on establishing a representative process flow diagram, defining reasonable operating assumptions, and identifying critical conversion steps that govern efficiency and yield. Detailed model calibration and optimization are ongoing.

Preliminary simulation results indicate that biomass-to-fuel conversion is achievable within realistic operating conditions. The model suggests that fuel production is strongly influenced by feedstock properties, conversion efficiency in the primary processing steps, and the overall energy integration within the system. While specific numerical results are still under development, insights into the key units that dominate material loss and energy demand have been obtained.

Based on the current findings of this project, biomass-derived rocket propellant production appears promising as a sustainable fuel pathway, especially in California's strict regulatory environment. However, further refinement of the ASPEN simulation and incorporation of economic evaluation are required to fully assess viability. Continued modeling efforts will focus on improving confidence in predicted performance and on finding opportunities for process optimization.

Introduction

Problem & Significance

The space industry is expanding as many new space-related opportunities arise. The space industry is expected to hit $1.8 trillion by 2035. In 2024, the US averaged 60 to 70 launches per week. The current RP-1 is made using fossil fuels, which have high Scope 1 emissions. Companies such as Amazon are adopting net-zero policies in their efforts to launch space satellites.

Supply & Demand Analysis

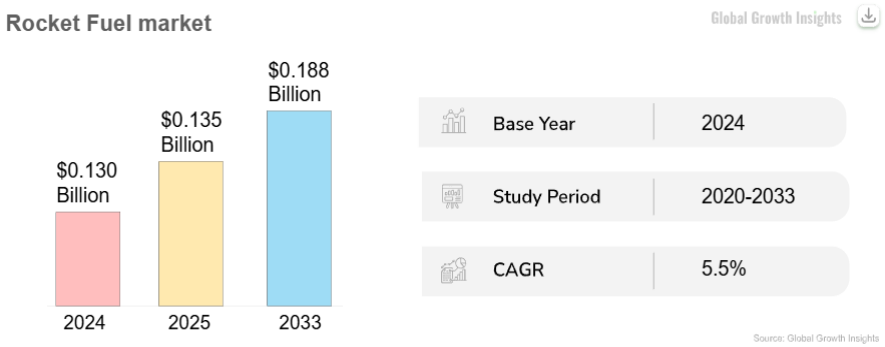


Figure 1: Projected Market Trend of the Rocket Fuel Market

While there is no supply-and-demand curve for RP-1, sufficient information is available on the current market and its projected future direction. The figure above shows the current and projected values of the rocket fuel market. It is currently around $0.130 billion and is projected to reach $0.188 billion. This shows a CAGR of 5.5%.

The space industry is expected to hit $1.8 trillion by 2035. It is currently at $630 billion. This indicates an average annual growth rate of 9%. Since the vehicle-launch industry is a niche compared to other petroleum industries, there is a limited supply of RP-1 due to the few suppliers. The other industries that heavily rely on petroleum-based fuels include land vehicles and aviation. Since these industries are more popular and in higher demand than the vehicle launch industry, more focus is placed on them.

Global Share

Table 1: Location, Global demand, and Rocket Fuel Used

Location	Global Rocket Fuel Demand	Rocket Fuel Used
US	35-40%	LOX/RP-1, LOX/methane
Asia-Pacific	30-35%	LOX/LH2
Europe	20-25%	LOX/LH2
Middle East & Africa	5-10%	LOX/RP-1

The above table shows the location, global demand, and rocket fuel used. This table is significant since it shows that the US is a leader in demand and uses RP-1. This is helpful since the plant location is easily decided.

In 2024, the US averaged 60 to 70 launches per week. By comparison, China launched only 60 missions in 2023.

Value Chain Analysis

Table 2: Value Chain Analysis Summary of Chosen Process

Stage	Value Added	Major Cost	Risk
Biomass Supply	Process begins	Cost of biomass	Supply variability
Pyrolysis	Conversion of biomass into bio-oil	Reactor heat, sand recycle	Tar formation
Combustion Loop	Self-sustaining heat	Air compression	Sand attrition

Quench & Condensation	Captures vapors and converts to liquid	Cooling	Phase separation issues
Hydrotreating	Converts bio-oil to hydrocarbons	H2, catalysts	Catalyst poisoning
Distillation	Separation	Reboiler energy	Coking
Hydrocracking	Produces fuel	H2, pressure, furnaces	Over-cracking
Hydrogen Loop	Supplies H2	Natural gas	Reforming downtime
Final Fuel	Final product	Distribution	Fossil fuel competition

The table above illustrates the value chain analysis of our chosen process. This breaks down each stage, describing the value added, major costs, and risks associated with each stage. This is important since it allows for further investigation into improving the process and where major problems can arise.

Material Prices

Table 3: Required Materials' Prices

Continuous Feedstock	**1st Run**	**Continuous Product**
Hybrid poplar wood chips	H2	RP-1

$0.03/bone dry pound delivered	$3/pound	$4.60/pound

The above table shows the material prices. The continuous feedstock is hybrid poplar wood chips, which cost $0.03 per bone-dry pound delivered. For the first run of our process, H2 is required, which costs $ 3 per pound. Our continuous product is RP-1, which can potentially be sold for $4.60/pound.

Sustainable RP-1 Price Calculation

- RP-1 from petroleum: $1.04/lb – $1.57/lb
- Sustainable aviation fuel (SAF) is around 2-5 times more expensive than fossil jet fuel
- Assuming the 2-5 times multiplier applies to rocket fuel
- Average RP-1 price: $1.31/lb
- Average of the multiplier: 3.5
- Sustainable RP-1 price: $4.60/lb

Since RP-1 is essentially a de-sulfurized kerosene, the assumption that the multiplier is similar is plausible. Since there is no information on the current price of sustainable RP-1, this is the best option.

Process Description

Competing Processes

Bio Propane

Bio-propane is a renewable form of propane produced through the hydrotreating of waste fats and bio-oil feedstocks. Its chemical properties are nearly indistinguishable from those of conventional propane, allowing it to be used in liquid-propane rocket engines with minimal modifications. Companies exploring bio-propane as a primary fuel include Orbex and BluShift Aerospace, driven largely by its sustainability benefits.

One of the key characteristics of bio-propane is its high energy density, which provides favorable mass efficiency for smaller launch systems. Its combustion is cleaner than that of kerosene-derived fuels, producing less soot and fewer sulfur oxides. Overall, its emissions profile is lower than its contemporaries.

This results in extended engine component life and reduces the environmental impact of launches. Another advantage is its storage behavior: bio-propane can be stored as a liquid under moderate pressure, avoiding the storage complexities associated with cryogenic fuels.

However, bio propane carries several disadvantages. Production costs are high due to the limited availability of renewable propane and the hydrotreating processes required to generate it. These feedstock and processing limitations constrain the large-scale adoption of biofuels, keeping their prices significantly higher than those of petroleum-derived fuels. Additionally, although it offers cleaner combustion, propane engines can still experience carbon deposition. These deposits increase maintenance needs and may offset some operational benefits.

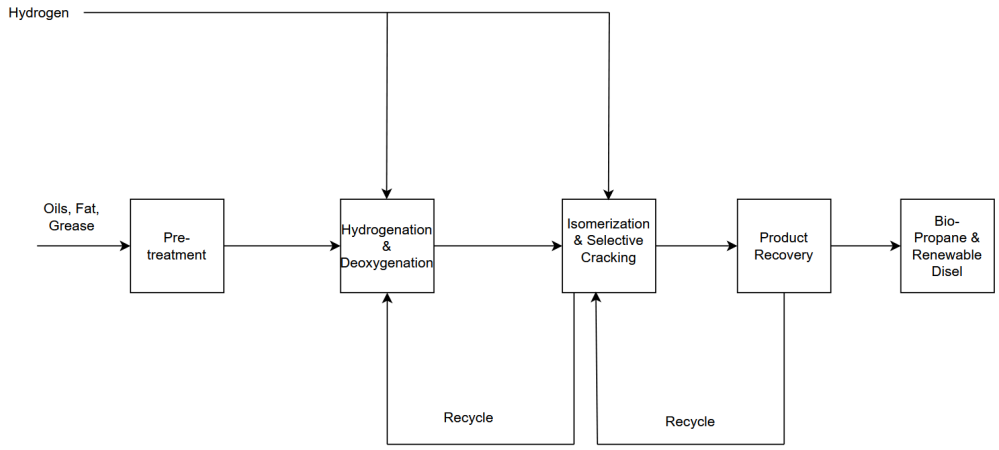


Figure 2: Propane Block Flow Diagram

Biomethane

Biomethane is a renewable, chemically identical substitute for fossil methane that can be produced through anaerobic digestion, biomass gasification, or synthetic-gas methanation. It has become an increasingly attractive rocket-fuel candidate as the industry explores high-performance methalox propulsion systems. Companies pursuing this include Interstellar Technologies and Air Water Hokkaido Inc., a collaboration between a space infrastructure company and an industrial gas producer.

One of biomethane's major advantages is that its production is commercially established on a global scale. Biogas upgrading to biomethane is common in the energy sector, and this maturity translates to stable supply chains. It offers cleaner combustion than kerosene, producing less particulate matter and reducing greenhouse gas impact when its renewable origin is considered. From a performance standpoint, methane-fueled rocket engines deliver high specific impulse and improved thermal efficiency, making them attractive for reusable launch systems and future deep-space missions.

Despite these advantages, biomethane presents significant engineering and logistical challenges. Methane engines are inherently more complex than kerosene engines, requiring precise cooling strategies, advanced turbomachinery, and tightly controlled combustion conditions. The fuel must also be stored cryogenically at low temperatures, necessitating insulated tanks, boil-off management systems, and additional ground infrastructure. Furthermore, producing biomethane from biomass often requires multiple upgrading steps, adding cost and technical complexity to the process.

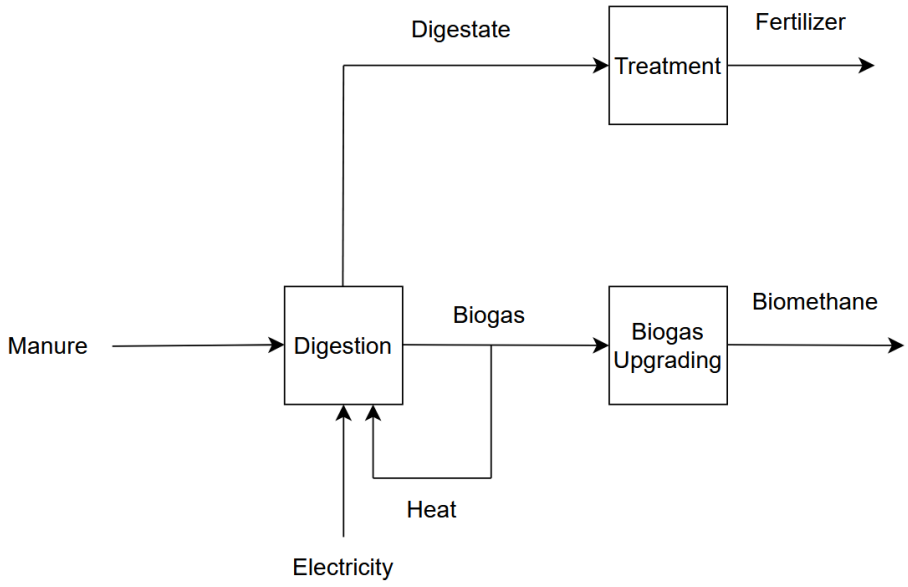


Figure 3: Methane Block Flow Diagram

Standard Crude-Oil

Crude-oil-derived rocket fuels, such as RP-1, remain the aerospace industry's most widely used hydrocarbon propellants. The fuel is produced through petroleum-refining processes that have been optimized over decades. Major users include SpaceX, Northrop Grumman, and the United Launch Alliance (ULA). The global refinery infrastructure supporting the fuel is mature, enabling a consistent supply at large volumes and at relatively low cost.

As mentioned previously, the largest benefit of crude-oil-derived RP-1 is their low production expense; it is significantly cheaper to produce than bio-based derivatives. Additionally, the process is highly standardized, with numerous documentations on its production. This is further supplemented by the decades of flight data on how the fuel behaves under high pressures and cryogenic temperatures. Chemically, the fuel is also rich in cyclic and branched alkanes, providing the high thermal stability required for regenerative cooling during launches. It also serves as a natural lubricant for the turbopumps and valves of the rocket.

However, there are several drawbacks, particularly in terms of long-term sustainability and environmental impact. Crude-derived RP-1 produces some of the highest carbon emissions among aerospace propellants, and their combustion generates soot and black carbon that can accumulate in the upper atmosphere. As global launch rates continue to rise, these environmental impacts cannot be ignored. Crude oil is also a finite resource; its deposits are limited within the Earth's crust. Additionally, producing RP-1 still requires deep desulfurization and extensive purification to remove contaminants such as sulfur, nickel, and vanadium. These processes increase energy demand and contribute to its environmental footprint further.

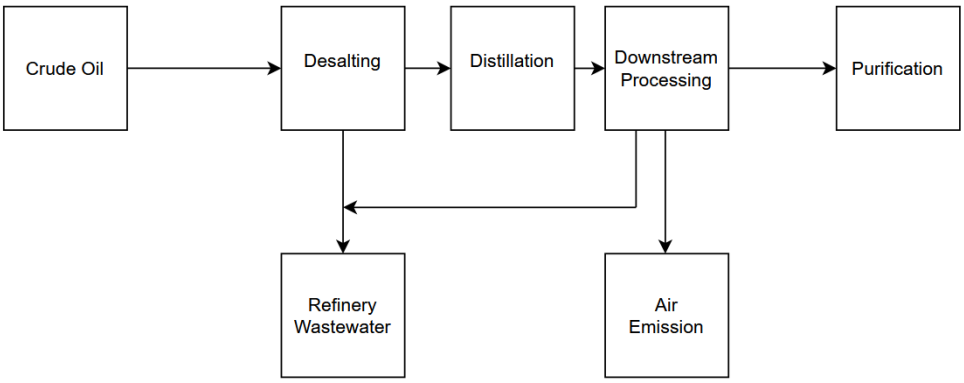


Figure 4: Standard Crude Oil Block Flow Diagram

Chosen Process

The chosen process of going from biomass to bio-oil has its complexities and constraints in the production process. The first stage of the process involves creating bio-oil from wood chips via fast pyrolysis. The process begins with the biomass entering a dryer to ensure it has zero to no moisture before entering the pyrolysis riser reactor, which operates at a nominal temperature of 500 °C to 930 °C. The effluent exits from the top, entering a cyclone that separates solids from the moisture. The moisture is carried up into stream 103 to undergo heat exchange and then heated to be quenched in the quench absorber. Following the outlet stream 106, the bio-oil then gets pressurized and heated before further processing. From the other outlet of the quench absorber stream 111, the vapor is compressed and then heat-exchanged to be fed as a fluidizing gas for the pyrolysis reactor. Returning to the cyclone, stream 115, a stream of solid, is fed into a combustor reactor at a temperature between 500 °C and 1000 °C, resulting in wet ash. From the wet ash stream, 116 gets sent to a cyclone to separate moisture from ash. The ash in stream 117 is still moist at a relatively low level. Stream 118, which is the moisture stream, gets heated up and exchanged with air. The air is used to feed the char combustor. The stream 124 then becomes flue gas, which heats the dryer and is exhausted through the exhaust stream.

Once bio-oil is produced, it must undergo hydrotreating to stabilize and upgrade the oil for use as a fuel. This is due to the nature of bio-oil being highly saturated with ~25% water, making it difficult to separate. Thus, it is unstable for long-term storage and not miscible in any conventional hydrocarbon-based fuels. The incoming bio-oil stream 200 is pressurized into a stream of H2, which is then heat-exchanged into the first hydrotreater, R-200, running at 240 °C and 2500 psig. The effluent from the hydrotreater, stream 221, is then heat-exchanged and heated in a furnace to approximately 300 °C. The heated stream is then fed into the second hydrotreater, which operates at 370 °C and 2015 psig. The reactor effluent is then heat-exchanged twice with an air cooler and further cooled in a heat exchanger using cooling water. Stream 207 is then fed into a high-pressure flash, which separates the gas from the oil. Following the liquid stream 208, it is then low-pressure flashed and heated in an exchanger in the bottom oil product. The overhead stream 213, which is then letting off fuel, will be fed into reformers. Following stream 215, the vapor stream is then fed into a pressure swing adsorption unit (PSA) to recover some H2 gas. Following the PSA stream, 217 is then fed a makeup H2 stream and subsequently compressed before being returned to the hydrotreaters.

The final stage of this design involves converting stable oil to RP-1. Once the oil from the bottoms is created, the oil is then fed to a debutanizer distillation. Where the distillate is fuel gas to be fed to heaters and reformers. The bottom product is then the stable oil that will be fed into a hydro desulfurizer, R-300. The reactor effluent is then fed into a flash drum to separate the gas from the stable oil. The tops, which are stream 307, will yield fuel gas and H2 for recycling in the hydro desulfurizer. The bottom, which is a stable oil, is then fed into a furnace to be heated before undergoing fractional distillation. From fractional distillation, a hard cut from the middle of the column is taken just before the dew point for the RP-1 kerosene (C12-C16). The distillate yields relatively low octane naphtha gas that can be sold as is or as a blend. The bottom product yields completely renewable and usable diesel that can also be sold.

Process Control

Fast Pyrolysis

The first phase, fast pyrolysis involves the conversion of woodchips into pyrolysis bio-oil through rapid heat and quenching. The objective is to reduce the moisture of incoming biomass to a suitable level before it enters the Pyrolysis Rise Reactor (R-100). With excess moisture, reaction temperature will be decreased and the water content of the bio-oil will increase, degrading its quality.

Biomass will enter the dryer (D-100) along hot flue gas from the Heat Exchanger (E-106). The flue gas will act as the drying medium. The dryer's temperature will be controlled, and the manipulated variable will be the flow rate of the flue gas. A temperature transmitter will be installed in the dryer with a control valve for the flue gas stream.

For fast pyrolysis to occur, the pyrolysis heat rise reactor will have to maintain a high heat environment. This prevents vapors from cracking into non-condensable gas and maximize bio-oil vapor yields. Entering the reactor are fluidizing gas, hot sand, and the dried bio mass. To maintain heat, a cascade controller will be utilized, this will monitor the temperature inside the reactor and control the flowrate of the hot sand.

Additionally, a pressure controller will be installed for the reactor. The manipulated variable will be the vent flowrate. This ensures oxygen ingress is prevented and reinforces operational stability.

After exiting the pyrolysis heat rise reactor (R-100), both solids and liquids must be condensed. The temperatures of the heat exchangers (E-100 and E-102) will be controlled. Cooling water will be supplied through the heat exchanger jackets and have their flowrates manipulated. A temperature sensor will be installed, signaling the controller to either increase or decrease its coolant flow based on the current conditions.

The stream enters the quench absorber (D-100). The volume of liquid exiting the drum must be controlled to prevent overflowing the gas compressor (K-100). Exiting the quench absorber, Stream-106, is the bottom stream and its flowrate will act as the manipulated variable.

The same stream will enter the decanter (D-100), where the organic bio-oil will be separated from the aqueous phase. An interface level controller will be installed to monitor the boundary between the two liquids and adjust the flow valves for each respectively.

Bio-Oil Upgrading

The second phase involves the conversion of bio-oil to standard oil. To do so requires the preparation of bio-oil for hydrotreatment via heating. With the raw bio-oil and hydrogen gas, a flow transmitter will be installed for Stream-204. Through this, a signal will be sent to a controller manipulating the flowrate of hydrogen gas by adjusting compression speed. This ensures the Reactor (R-200) runs without coke formation and prevents excessive heat buildup.

To further safeguard against heat, cascade control will be implemented for Reactor (R-200) to manage temperature. A thermocouple will measure the temperature of the reactor's catalyst bed and its signal wll be sent to a controller maintaining the cooling water flow to the reactor's jacket. Any excess heat will be absorbed and keep the temperature at desired temperatures.

Afterwards, the effluent will exit the reactor and pass through the heat exchanger (E-200). The temperature will be controlled through a feedback control loop that will manipulate the flowrate of the cooling water through the heat exchanger.

With the cooled mixture, the stream enters the High-Pressure Flash Drum. High levels of pressure are required to direct the off-Gas (hydrogen) towards the PSA units. A pressure controller will manipulate a control valve on the gas outlet stream (Stream-209) to maintain constant pressure. An additional feature to the drum will be a liquid level controller, preventing both the gases and liquids entering their undesignated streams. The manipulated variable will be flow of the bottom stream (Stream-208). There will also be an interface controller for the wastewater stream to detect oil and water; a valve will be manipulated to release the water from the drum.

The liquid product from the High-Pressure Flash Drum will then enter the Low-Pressure Flash Drum (D-200B). Any remaining gases will be eliminated and vaporized. Pressure will be controlled via a vent valve for the effluent gas stream. Additionally, liquid level will be controlled through flowrate manipulation of the product stream (Stream-214).

DC-200, the decanter, will separate the upgraded bio-oil from any remaining water. An interface level controller will measure the bio-oil and water boundary; the flowrate of the effluent (Wastewater 2) will be maintained to achieve a constant water level. Monitoring the liquid volume in the decanter will be a level controller. The flowrate of the product stream will be manipulated by a flow valve to send the stable bio-oil to the centrifugal pump, the next phase of the process.

Fractionation

The third and final phase will require the upgraded bio-oil to enter the debutanizer column (C-300), followed by a reboiler and fractionation column to produce the desired products. To manage feed to the column, a feed transmitter will be installed.

The primary function of the debutanizer is to increase the flashpoint of the liquid mixture and remove any remaining gases. To ensure a regulated introduction of liquid into the column, a dedicated flow control valve (FV-300) and swing check valve will be utilized to prevent any disturbances in the Vapor-Liquid Equilibrium and pressure. Connected to FV-300 will be a temperature element and temperature transmitter monitoring the column's upper tray. The transmitter will be supplemented with high and low temperature alarms, which will alert the plant of thermal deviations. If tray temperature rises, the reflux valve (FV-300) will open to introduce cold liquid into the column. Directly from the overhead of tower, is a pressure relief valve to circumvent the tank from rupturing.

Additionally, the reflux drum will have a level indicator to manipulate the butanes exiting via the product valve (LV-300), followed by both a level controller and high/low liquid level alarms. A flow transmitter on the line will report the yields of the stream.

To help further manage the VLE, the reflux drum will include a pressure transmitter, pressure high/low level alarms and a pressure controller. The transmitter will be connected to the pressure valve (PV-301) of the vapor outlet. This will purge both the methane and ethane that have accumulated in the system.

At the base of the debutanizer column is a reboiler. The distillate of the tower will be sent to the reboiler to be heated. Entering the reboiler will be the heating medium, steam, which will be managed by a flow controller, a flow transmitter, and valve (FV-301). The liquid height will be monitored through the installed level transmitter, level high/low alarms, and a controller. The controller will be connected to the product valve (LV-300) heading to the fractionation column. This is to prevent any mechanical failure caused by the overheating of the reboiler tubes. Joining the level valve (LV-301) will be a swing check valve to direct liquid flow.

The second column is the fractionation column (C-301), where the mixture will be separated into three liquid products: RP-1, Naphtha, and Diesel. The previous feed from the debutanizer column will enter through Stream-304.

At the top section of the column, the overhead is condensed into a liquid. Pressure maintenance is the primary objective and will be managed by a pressure controller. This controller will adjust the fan speeds of the condenser. Liquid management will be performed through a level controller on the reflux drum. The drum's Naphtha level is monitored, and the discharge flow rate is manipulated. A flow controller is used for the reflux to maintain a constant reflux ratio and prevent kerosene molecules from remaining in the Naphtha.

At the side-draw point, a temperature controller will monitor the nearby tray in the column. The controller will provide a setpoint for flow controller on Stream-306. The system will automatically adjust the flowrate of the liquid based on tray temperature, preventing RP-1 from being contaminated from Naphtha or Diesel.

The heaviest hydrocarbon is the Diesel product exiting as Stream-307. A level controller at the base of the fractionation column will monitor the liquid level and manipulate the product valve on Stream-307.

The column is managed by the reboiler with its heat managed via a cascade loop. A temperature controller will be installed on the stripping-section tray and provide a setpoint for a steam flow controller. The regulated steam flow will send the lighter RP-1 molecules back up the column, leaving only Diesel to exit the bottom.

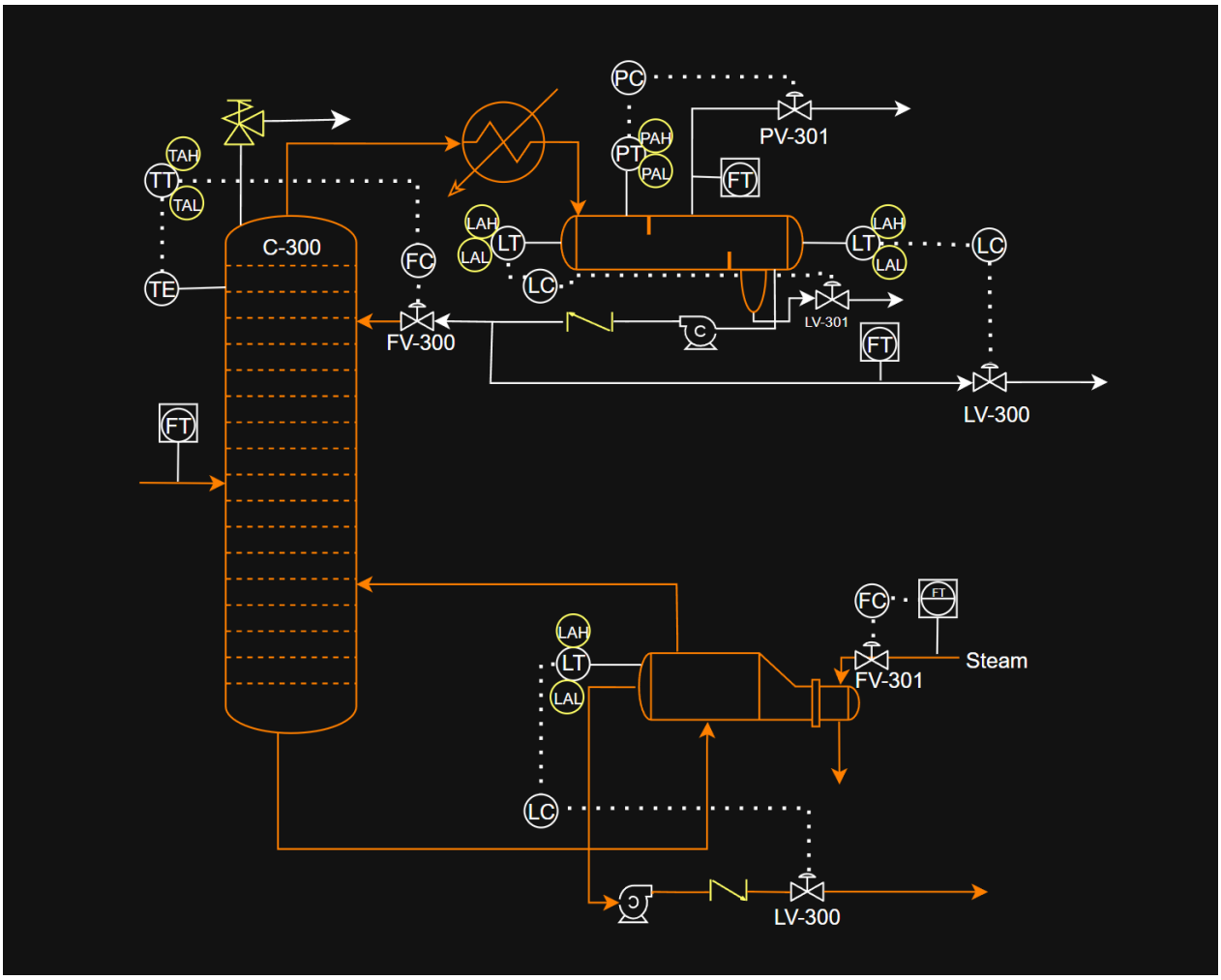


Figure 5: Debutanizer Column Control Loop

Environmental & Safety Analysis

Since the proposed facility is located in California, all emissions and waste streams must comply with environmental regulations enforced by the United States Environmental Protection Agency (EPA) and the California Air Resources Board (CARB). Air emissions from industrial facilities are regulated under the Clean Air Act, while liquid discharges are governed by the Clean Water Act through the National Pollutant Discharge Elimination System (NPDES). These regulations limit emissions of pollutants such as sulfur oxides (SOx), nitrogen oxides (NOx), particulate matter, and volatile organic compounds (VOCs), as well as contaminants present in wastewater.

The proposed fast pyrolysis, bio-oil upgrading, and fractionation processes generate gaseous, liquid, and solid by-products that must be properly managed to minimize environmental impacts. To ensure regulatory compliance, several emission control and waste treatment systems are incorporated into the facility design.

The fast pyrolysis reactor generates a non-condensable gas stream composed primarily of nitrogen, oxygen, carbon dioxide, hydrogen, water vapor, and trace amounts of sulfur dioxide (SO_2) and nitrogen oxides (NOx). In addition, entrained ash particles from the biomass feedstock may be present in the vent stream. Without proper treatment, these emissions could contribute to atmospheric particulate pollution and acid gas formation.

Industrial combustion and thermal processing equipment operating under the Clean Air Act typically must limit NOx emissions to approximately 30–100 ppmv, depending on the technology and permit requirements. In California, regional air quality districts frequently enforce more stringent limits near 30 ppmv for permitted equipment. Particulate matter emissions are commonly restricted to approximately 0.03–0.1 lb/MMBtu of heat input.

To control particulate emissions, the pyrolysis gas stream should first pass through cyclone separators followed by baghouse filtration systems capable of removing more than 99% of entrained ash particles. Acid gases such as SO_2 can be reduced using wet or dry scrubbers that neutralize sulfur compounds prior to atmospheric release. Combustible gases, including hydrogen and light hydrocarbons, should be routed to a flare or thermal oxidizer where controlled combustion converts them primarily to carbon dioxide and water while achieving high destruction efficiency. Pressure relief valves on reactors and process vessels should also be connected to the flare header to prevent uncontrolled hydrocarbon release during upset conditions. Continuous emissions monitoring systems can be installed to measure NOx, SO_2, and particulate concentrations and verify compliance with permit limits established by CARB and the EPA.

The bio-oil upgrading unit generates gaseous streams containing hydrogen, carbon dioxide, hydrocarbons ranging from C_1 to C_{16}, and various oxygenated organic compounds. Many of these compounds are classified as volatile organic compounds (VOCs), which are regulated due to their role in the formation of ground-level ozone and photochemical smog.

Industrial control devices operating under Clean Air Act requirements typically must achieve VOC destruction or removal efficiencies between 90 and 98%. In California, air quality regulations frequently require final VOC concentrations in vent streams to be below approximately 10–20 ppmv after treatment, depending on the permitting authority.

To control VOC emissions, the upgrading system should incorporate closed-loop vapor recovery systems that capture hydrocarbon vapors and recycle them as process fuel or feedstock within the plant. Non-condensable vent streams and pressure relief discharges should be routed to a flare or thermal oxidizer to ensure effective destruction of VOCs and combustible gases. In addition, a Leak Detection and Repair (LDAR) program should be implemented to monitor valves, pumps, compressors, and flanges for fugitive hydrocarbon emissions. Routine inspection using portable gas analyzers allows leaks to be identified and repaired promptly, minimizing uncontrolled VOC releases.

Wastewater generated during the bio-oil upgrading process may contain dissolved hydrocarbons and oxygenated organic compounds that contribute to biological oxygen demand (BOD) and chemical oxygen demand (COD). If released untreated, these compounds could deplete dissolved oxygen in receiving waters and negatively impact aquatic ecosystems.

Under the Clean Water Act and NPDES discharge permits, industrial wastewater streams commonly must meet limits of approximately 30 mg/L BOD, 30 mg/L total suspended solids (TSS), and 10–15 mg/L oil and grease prior to discharge.

Wastewater should therefore be collected and routed to a treatment system designed to remove hydrocarbons and organic contaminants. Initial oil-water separation can remove free hydrocarbons, followed by pH neutralization and biological treatment processes that reduce organic loading through microbial degradation. Final polishing steps, such as activated carbon adsorption, may be used to remove residual oxygenated compounds before discharge. Equipment wash water and scrubbing liquids should also be directed to this treatment system. In addition, the plant cooling system should operate as a closed-loop recirculating system to minimize cooling water blowdown and reduce overall wastewater generation.

The fast pyrolysis process produces ash as a solid by-product of biomass conversion. Although biomass ash is typically less hazardous than coal ash, it may contain fine particulates and trace metals that require proper handling. Cyclone separators and baghouse filters capture the majority of ash particles, preventing their release to the atmosphere. The collected ash can then be stored in sealed containers and transported to approved disposal facilities. If chemical analysis confirms that contaminant concentrations are within acceptable limits, the ash may potentially be reused as a soil amendment or construction material, reducing overall waste generation.

Additional solid wastes may include spent filtration media, wastewater treatment sludge, and deactivated process catalysts. Catalysts used in bio-oil upgrading reactors may gradually lose activity due to coke deposition or metal accumulation. These materials should be collected and either regenerated by specialized off-site facilities or disposed of in accordance with hazardous waste management regulations. Proper storage in labeled containers and adherence to approved waste management practices are required prior to disposal.

The fractionation section produces an overhead vapor stream containing hydrogen, light hydrocarbons (C_1–C_{16}), water vapor, and oxygenated organic compounds. Because these species contribute to VOC emissions, the vapor stream should first pass through condensers that recover condensable hydrocarbons and oxygenates for reuse within the process. Remaining non-condensable gases should be routed to a flare or thermal oxidizer capable of achieving VOC destruction efficiencies above 90–98%, ensuring compliance with federal and California air quality regulations. Heat recovery from flue gases may also be implemented to improve overall plant energy efficiency and reduce fuel consumption.

By integrating multiple emission control and waste management technologies—including cyclone separators, baghouse filtration, scrubbers, condensers, vapor recovery systems, flare or thermal oxidizer systems, wastewater treatment units, and leak detection programs—the facility can significantly reduce environmental impacts while complying with regulatory requirements. These design measures ensure that emissions of particulate matter, VOCs, acid gases, and wastewater contaminants remain within permitted limits while supporting responsible environmental stewardship.

Ethical & Legal Responsibilities

The facility handles flammable gases, high-temperature process streams, and pressurized equipment. Plant operation must follow strict ethical and legal safety responsibilities to protect workers, the surrounding community, and the environment. Worker safety requirements in the United States are enforced by the Occupational Safety and Health Administration (OSHA), particularly through the Process Safety Management Standard for facilities handling highly hazardous chemicals. These regulations require employers to identify process hazards, maintain safe operating procedures, provide worker training, and implement engineering controls that reduce the risk of fires, explosions, or toxic releases. In the proposed pyrolysis and bio-oil upgrading facility, employees may be exposed to hazards such as high temperatures, pressurized hydrogen streams, combustible hydrocarbon vapors, and rotating mechanical equipment. To

mitigate these risks, all personnel must receive comprehensive safety training covering chemical hazards, safe operating procedures, emergency response protocols, and proper equipment handling before working in the process area. Personal protective equipment such as flame-resistant clothing, chemical-resistant gloves, safety goggles, hard hats, and steel-toe boots should be required when working near operating units. In areas where hydrogen or hydrocarbon vapors may accumulate, portable gas detectors may also be used to alert workers to potentially dangerous concentrations.

Beyond worker safety, plant designers and operators also have an ethical obligation to protect the surrounding community and environment from accidental releases of hazardous materials. Public safety requirements are enforced primarily by the United States Environmental Protection Agency under environmental legislation such as the Clean Air Act and the Clean Water Act. These regulations require industrial facilities to limit air emissions, treat wastewater prior to discharge, and implement measures to prevent uncontrolled releases of pollutants. In addition to regulatory compliance, ethical engineering practice requires that process systems be designed with multiple layers of protection to minimize the likelihood and consequences of accidental releases. For example, equipment handling hydrogen and hydrocarbon vapors should be constructed to appropriate pressure ratings, regularly inspected for leaks, and equipped with automatic isolation valves and containment systems that prevent the spread of hazardous materials outside the plant boundary.

Emergency response preparedness is another critical component of responsible plant design. Because the process involves flammable gases and elevated temperatures, emergency shutdown systems (ESD) must be incorporated into the process control architecture. These systems automatically shut down critical equipment if unsafe operating conditions are detected, such as excessive reactor temperature, high pressure, or loss of cooling. Alarm systems connected to distributed control systems can notify operators when process variables exceed safe operating limits, allowing corrective action to be taken before hazardous conditions develop. Safety interlocks may also be installed to prevent equipment from operating outside safe conditions; for example, interlocks can prevent a reactor feed pump from starting unless proper cooling and inert gas flow are established. In addition, pressure relief devices routed to flare systems provide a final layer of protection by safely venting excess pressure during abnormal operating conditions.

Finally, emergency response planning must include procedures for fire suppression, evacuation, and coordination with local emergency services. Operators should be trained in emergency drills that simulate potential incidents such as gas leaks, fires, or equipment failures. Clear evacuation routes, safety signage, and emergency communication systems must be established throughout the facility to ensure that personnel can respond quickly and safely in the event of an incident. By implementing robust safety training programs, requiring appropriate personal protective equipment, and integrating automated safety systems such as alarms, shutdown systems, and interlocks, the facility can meet both legal regulatory requirements and the ethical responsibility of protecting workers, the public, and the environment

Safety- Process Definition

The proposed process converts lignocellulosic biomass, specifically woodchips, into a liquid hydrocarbon fuel suitable for use as rocket propellant similar to RP-1. The process begins with the preparation and fast pyrolysis of biomass feedstock, where woodchips are rapidly heated in the absence of oxygen to produce bio-oil, non-condensable gases, and solid char or ash. The bio-oil is then routed to an upgrading section where catalytic reactions remove oxygenated compounds and convert the product into stable hydrocarbons. Finally, the upgraded product is sent to a fractionation system that separates hydrocarbons into different boiling ranges to isolate the desired fuel fraction appropriate for rocket propellant

applications. Major unit operations in the process therefore include the fast pyrolysis reactor, gas-solid separation equipment such as cyclones or filters, catalytic upgrading reactors, heat exchangers, and distillation or fractionation columns used for product purification and separation.

The facility operates over a wide range of process conditions depending on the unit operation. Process stream temperatures range from approximately 77°F to 1129.9°F, with the highest temperatures associated with the hot sand circulating in the pyrolysis reactor used to rapidly heat the biomass feedstock. Pressures across the plant range from approximately 14.7 psi to 1450.4 psi, with the highest pressures occurring in the catalytic upgrading reactors where hydrogenation reactions are carried out. Flow rates vary significantly throughout the facility, ranging from relatively small recycle streams to major process streams exceeding hundreds of thousands of pounds per hour. Streams within the process exist in both gas and liquid phases depending on temperature, pressure, and composition. The primary chemicals present in the process include biomass feedstock (woodchips), hydrogen used for upgrading reactions, carbon dioxide, nitrogen, water, light hydrocarbons ranging from C_1–C_{16}, oxygenated organic compounds derived from pyrolysis oil, and the final hydrocarbon fuel product. These materials are present in various process vessels and pipelines throughout the plant, with the largest inventories located in the pyrolysis reactor system, upgrading reactors, and product fractionation columns.

Because the process involves high temperatures, elevated pressures, flammable hydrocarbons, and hydrogen gas, the plant has been designed with multiple layers of safety protection including pressure relief systems, process control loops, gas monitoring, and emergency shutdown capabilities to minimize the risk of fire, explosion, or accidental release.

Chemical Safety & OSHA PSM Requirements

Chemical Hazard Table

Chemical	Hazard Type	Flow Rate (lb/h)	Stream/Location	OSHA Classification / Notes
Hydrogen	Highly flammable gas	~6,500	Hydrotreater & recycle loop	Covered under OSHA PSM flammable gas category if inventory >10,000 lb
Carbon Monoxide	Toxic, flammable	~34,900	Pyrolysis gas streams	Covered under OSHA PSM flammable gas category

Methane	Highly flammable gas	~2,400	Fuel gas & reformer streams	Covered under OSHA PSM flammable gas category
Ethylene	Flammable gas	~2,100	Pyrolysis gas	Covered under OSHA PSM flammable gas category
Sulfur Dioxide	Toxic, corrosive gas	~660–1,175	Combustion flue gas	Regulated under OSHA hazardous air contaminant exposure limits
Nitrogen Dioxide	Toxic, oxidizing gas	~145–278	Combustion flue gas	Regulated under OSHA exposure limits for respiratory toxins
Acetone	Flammable liquid	~2,834	Bio-oil stream	Covered under OSHA flammable liquid hazard category
Acetic Acid	Corrosive, irritant	~32,800	Bio-oil stream	Regulated under OSHA corrosive chemical exposure standards
Furfural	Toxic, flammable organic	~8,600	Bio-oil stream	Regulated under OSHA toxic organic chemical exposure limits
Cresols / Phenolic Compounds	Toxic, combustible	~11,900	Bio-oil	Regulated under OSHA toxic organic chemical exposure limits
Guaiacol	Toxic organic compound	~81,800	Bio-oil fraction	Regulated under OSHA hazardous organic exposure standards

Levoglucosan	Combustible organic solid	~13,000	Bio-oil stream	Considered combustible organic material with fire hazard
Carbon Dioxide	Asphyxiant gas	~35,800	Pyrolysis & combustion gas	Regulated under OSHA confined space & exposure standards
RP-1 Rocket Fuel	Flammable hydrocarbon liquid	~35,000	Distillation section	OSHA PSM flammable liquid category if storage >10,000 lb; high fire/explosion risk

Operational & Process Hazards

Process Hazard Table

Hazard Type	Description	Causes	Consequences	Safeguards
Pressure Hazard	Potential for overpressure in hydrotreating reactors	Hydrogen compressor malfunction; Blocked outlet line; Excess hydrogen feed; Thermal expansion of fluids	Reactor vessel rupture; Hydrogen fire or explosion; Equipment damage or plant shutdown	Alarms; Pressure relief valves; Regular inspections
Temperature Hazard	High temperature streams in pyrolysis reactor	Reactor operates at ~500–900°C; Hot sand circulation	Severe burn risk; Equipment damage/overheating; Damage to piping	Thermal insulation; Restricted access; Temperature monitoring

Flow Hazard	Loss of flow to pyrolysis reactor	Biomass feed blockage; Conveyor belt malfunction	Incomplete pyrolysis; Temperature instability	Flow sensors on feed; Reactor shutdown if feed stops

HAZOP Analysis

C-300 - Debutanizer

Intention- Separate lighter hydrocarbons (C1-C4) from heavier ones in the upgraded bio-oil stream to ensure downstream products meet combustion specifications for rocket fuel production

Guide Word	Deviation	Cause	Consequences and Safeguards
LESS OF	Feed flow	Feed pump failure, upstream blockage, control valve partially closed	Reduced vapor generation and poor separation Safeguards: Flow alarms, upstream monitoring
NO/NONE	No feed flow to columns	Feed pump failure, upstream blockage	Column runs dry leading to poor separation and unstable operation. Safeguards: Flow alarms, standby feed pump, operator monitoring
MORE OF	Feed flow	Upstream pump overspeed, control valve malfunction	Column flooding due to excess liquid, reduced tray efficiency, overall poor separation. Safeguards: Feed flow control loop, high column pressure alarms, level monitoring
LESS OF	Low reflux flow	Reflux pump failure or valve malfunction	Reduced separation efficiency leading to

			carryover of heavier hydrocarbons in distillate stream. Safeguards: Reflux flow alarms, backup reflux pump, automatic flow control
NO/NONE	No reflux	Reflux pump failure or valve failure	Column loses separation capability and light hydrocarbons may contaminate the bottoms product Safeguards: Pump redundancy, flow alarms
MORE OF	High reflux flow	Control valve malfunction or condenser overcooling	Increased liquid level in column, possible pump cavitation and tray flooding Safeguards: Level alarms, control valve tuning, relief system
MORE OF	High pressure in column	Condenser failure or blocked overhead line	Risk of column overpressure which may lead to equipment damage or release of hydrocarbons Safeguards: Pressure control loops and alarms
LESS OF	Low column pressure	Excessive condenser duty or vent leak	Increased vaporization causing poor separation and unstable column operation Safeguards: Pressure control loops and alarms
MORE OF	High column temperature	Reboiler overheating or excess heat input somewhere	Thermal degradation of hydrocarbons and increased pressure in the column

			Safeguards: Reboiler temp control, and alarms
LESS OF	Low column temperature	Insufficient reboiler heat duty	Incomplete vaporization leading to poor separation and off-spec product Safeguards: Reboiler temp control and alarms
EARLY/LATE	Delayed control response	Slow valve actuation or controller tunning issues	Unstable column temperature and pressure profiles, poor separation Safeguards: Proper controller tuning and maintenance
MORE OF	High liquid level in column	Blocked bottoms line or pump failure	Risk of flooding within column trays and reduced separation Safeguards: Level sensors, alarms, automatic shutdown

A Hazard and Operability (HAZOP) analysis was conducted for the debutanizer column, C-300, to evaluate potential process deviations that could compromise safe and efficient operation. This unit is critical in ensuring that the downstream products meet specification needed for rocket fuel, therefore being able to mitigate any instance of variance from our setpoints is essential to the plant. The HAZOP framework was applied using standard industry guide words such as “Less OF,” “MORE OF,” and “NO/NONE” to identify the deviation deviations in major operating parameters including feed flowrate, reflux, pressure, temperature, and liquid level. For each deviation, as is standard, possible causes are identified. Examples of these deviations include; pump failures, valve malfunctions, or heat exchange issues. Then, the consequences of such deviations were considered.

The safegaurds included for each deviation were selected to address the specific risks identified in the analysis. These safeguards include instrumentation such as flow, pressure, and level alarms, along with control systems like feedback loops and automatic shutdowns, as well as equipment-based solutions such as backup pumps. Each safeguard plays a role in either preventing the deviation, detecting it early, or reducing its impact on the process. By connecting each deviation to realistic causes and practical safeguards, the analysis shows how the column can continue to operate safely even if issues arise. There are an endless number of deviations that could arise, but the ones presented are the major ones to consider and serve the purpose of this report.

Economic Analysis

ISBL

The inside battery limits (ISBL) include all the major equipment cost. The ISBL also considers bulk items like pipe, valves, instruments, and other items that aren't necessarily in the aspen flow sheet. From the Aspen simulation the total capital cost came out to be roughly $29,000,896 and this value is using data from the year 2022 therefore a escalation factor was used to get the 2026 value and that value came out to be $28,598,411. The larger equipment cost can be attributed to the columns, the rise reactor and the hydrotreater unit. Using the capital cost and the appropriate lang factor of our process the following is obtained where Ce is cost of equipment and F is the lang factor.

$$ISBL = Ce * F$$

$$ISBL = \$28{,}598{,}411 * 3.63 = \$103{,}812{,}231.9$$

The lang factor of 3.63 was obtained from (Towler & Sinnott, 2013)

Table 5: Equipment Pricing

Equipment	Price
Dryer-100	$3,431,400.00
R-200	$2,400,000.00
E-400	$29,000.00
R-300 (combustor)	$900,000.00
E-401	$100,600.00
Sep-400	$43,400.00
ST-400	$2,282,600.00
E-300	$374,013.72
E-301	$23,700.00
E-302	$49,677.55
B4 heater	$20,700.00
B5 pump	$242,500.00
B2 multi stage comp	$1,535,800.00
B3 hydrotreater	$315,900.00
B7 heater	$35,300.00
B8 flash 3	$131,700.00
B9 sep PSA	$29,300.00
B10 Flash 2	$56,700.00
B14 pump	$29,200.00
B12 debutnizer	$190,800.00
B15 pump	$9,800.00
B13	$771,100.00
B21	$984,100.00
B16 pump	$7,600.00
Rize reactor	$5,000,000.00
E-302	$29,222.09
E-300	$220,008.07
R-300 Char combustor	$900,000.00
R200	$2,400,000.00
R-200 E-101	$160,132.45
Reboiler B13	$27,911.82
Reboiler B12	$39,294.12
Reboiler B21	$23,892.46
Conndesor B13	$25,200.00
Condensor B12	$19,500.00
Condensor B21	$33,200.00

OSBL

The OSBL is the cost in the investment that the company must make for the site's infrasctructure, which include things like power lines, power generation plnats, genrators, cooling towers, and so on. The OSBL is generally calculated as a proportion of the ISBL cost. The cost percentage cost for a petrochemical plant is somewhere around 40% , and because our plant is biomass to fuel the assumption that was made is that it's the same percentage. Giving us the total OSBL as the following

$$OSBL = .40 * \$103{,}812{,}231.9 = \$41{,}524{,}892.76$$

Fixed Capital Cost

Taking the total of both battery limits ISBL and OSBL, the fixed capital cost was calculated to be

$$FCC = \$103{,}812{,}231.9 + \$\,41{,}524{,}892.76 = \$145{,}337{,}124.66$$

Contingency

Contingency charges allow for the project to budget for the variations of unexpected cost that come from changes in the project. Changes like in material costs. Usually a contingency charge is about 10% of the ISBL cost in the project.

$$Contingency = .10 * ISBL = \$103{,}812{,}231.9 * .10 = \$10{,}381{,}223.19$$

ECC (Engineering Cost)

The engineering cost considers the engineering services that are used to design the project that include design of pipping systems, control systems, plant layout, and scale modeling. Engineering cost can typically range between 10% to 30%. A value of 10% was chosen for our process.

$$ECC = .1 * ISBL = \$10{,}381{,}223.19$$

Working Capital

The working capital is the amount of money needed for the initial startup of the plant to run initially. This cost covers raw material inventory and other costs of operations. The working capital is typically 5% to 30% of the ISBL. For this process given the large amount of feed and only really having one raw material the percentage chose is 15 percent of the ISBL.

$$WC = .15 * ISBL = \$15{,}571{,}834.785$$

Total Capital Cost

The total capital cost is calculated in the following manner.

$$TCC = ISBL + OSBL + ECC + Contigency + WC = 181{,}671{,}405.9$$

Variable Cost of Production

The variable cost of production or VCOP is the cost of materials, utilities, and consumables used during the production. For this project, there is only one raw material which is Wood chips. Assuming that a operating year is 8000 hours the raw material cost came out to be $79,200,000. The utilities used in the process are electricity, low-pressure steam, cooling water, Green hydrogen, CoMo catalyst,

wastewater treatment, and Fuel fire making the total utility cost $MM107.7 per year. Adding the raw material cost, utilities cost, waste water treatment, and subtracting the profit made from our side products the VCOP comes out to be.

$$VCOP = 79200000 - (36505.37 - 10823.38) + 107742756.4 = \$MM186.9$$

Fixed Cost of Production

Fixed cost of production involves costs that do not vary with the rate of production. The first involves labor costs that have 5 operators, 4 shifts, and paying them $120,000 dollars each. Additionally, there should be a supervisor costing 25% of the operating labor. The direct overhead was taken as 45% of operating labor, The Labor and maintenance was taken at 65% and tax and insurance was taken at 2%

(1)Cost of shift operators= (5*4*$120,000)/1000000 = $2.4MM/year

(2)Supervision Costs= 25%*$2.4MM/year= $.60MM/year

(3)Direct Overhead cost= 45%*(2.4+.60) = $1.35MM/year

(4)Maintenance = 3% of ISBL = $6.48MM/year

(5)Labor and Maintenance = Number of shifts + supervision + Direct overhead + Maintenance = $10.28MM/year

FCOP = sum (1-5) = $21.11 MM/year

Revenue

Revenue accounts for all the income from the sale of main product and side product. The main product being Rp-1 and the side products being Naphtha and Diesel, and the total revenue came to be.

RP-1 Profit ($/year)	$484,737,600.00
Diesel Profit ($/year)	$156,209,042.41
Nahptha Profit ($/year)	$26,204,320.00

$$Total\ Revenue = \frac{\$770.21MM}{year}$$

NPV

Net present value is the sum of the present values of the future cash flows and is a method to see id the process is a smart choice to go with. The first two years account for the construction of the plant and each year account for half the total investment of the plant. Starting by year 3, Production begins generating profit which is calculated using FCOP and VCOP alongside the revenue. A 7-year MACRS was used to analyze the economics of this process. Depreciation was founded by multiplying the MACRS values with the total fixed capital of the plant.

$$Gross\ Profit = Main\ Product\ Revenues - CCOP$$

$$Depreciation = MACRS * (Total\ fixed\ capital\ cost)$$

The cash flow for each specific year is then gross profit subtracted by the amount of taxes paid. The discount factor is utilized to calculate the modified present value of cash flow.

$$Cash\ Flow = Taxable\ Income - Cash\ Flow$$

$$Discount\ Factor = 1/(1 + interest)^2$$

$$Present\ value\ of\ Cash\ Flow = Cash\ Flow * Discount\ Factor$$

Finally the NPV can be calculated by using the equation below, where the cash flow in each year is added and affected by the interest rate.

$$NPV = \sum \frac{CF_n}{(1+i)^n}$$

After calculating the final NPV after 21 years in the plant operation, the NPV values for 7 year MACRS was determined to be $3426MM meaning we have a net profit after 21 years of operation. Additionally, the DCROF was calculated to be 44 percent.

Gross margin

Gross Margin is the sales revenue is what the company keeps after being affected by the direct costs when producing the product. Gross Margin can be found by the sum of product and by the by product revenues subtracted by raw material costs. Given that our process generates large amounts of money its safe to say the gross margin is roughly $1350MM in profits.

Sensitivity Analysis

The sensitivity Analysis allows for the examination of uncertainties from changes of raw material price, sales price and other things that might affect the NPV. A range of error between -30% to 30% is introduced to each factor, and all the factors are ISBL, Revenue, VCOP and FCOP, and this sensitivity analysis was done with the 7-year MACRS and the final NPV was being examined through the analysis.

Table 6: 7 years MACRS

7 year MACRS								
Plant life	Pre tax cash flow	Depreciation	Taxable income	Tax	CF	FD	PV of CF	NPV
1	-385		-385	0	-385	0.87	-337.81	-337.81
2	-385		-385	0	-385	0.76	296.32	-634.13
3	270	23.73	246.81	0	270.5	0.67	182.61	-451.52
4	562	40.67	521.52	74.04	488.15	0.59	289.02	-162.49

5	562	29.1	533.15	156.45	405.74	0.52	210.73	48.23
6	562	20.7	541.45	159.94	402.25	0.45	183.26	231.49
7	562	14.8	547.37	162.43	399.76	0.39	159.76	391.25
8	562	14.81	547.38	164.21	397.98	0.35	139.51	530.77
9	562	14.83	547.37	164.21	397.98	0.31	122.38	653.15
10	562	7.41	554.79	164.21	397.98	0.27	107.35	760.51
11	562	166.1	396.10	166.43	395.76	0.24	93.64	854.15
12	562	0	562	118.83	443.36	0.21	92.02	946.18
13	562	0	562	168.66	393.54	0.18	71.65	1017.83
14	562	0	562	168.66	393.54	0.16	62.85	1080.68
15	562	0	562	168.66	393.54	0.14	55.13	1135.81
16	562	0	562	168.66	393.54	0.12	48.36	1184.18
17	562	0	562	168.66	393.54	0.11	42.42	1226.61
18	562	0	562	168.66	393.54	0.09	37.21	1263.82
19	562	0	562	168.66	393.54	0.08	32.64	1296.46
20	562	0	562	168.66	393.54	0.07	28.63	1325.1
21	562	0	562	168.66	393.54	0.06	25.11	1350.22

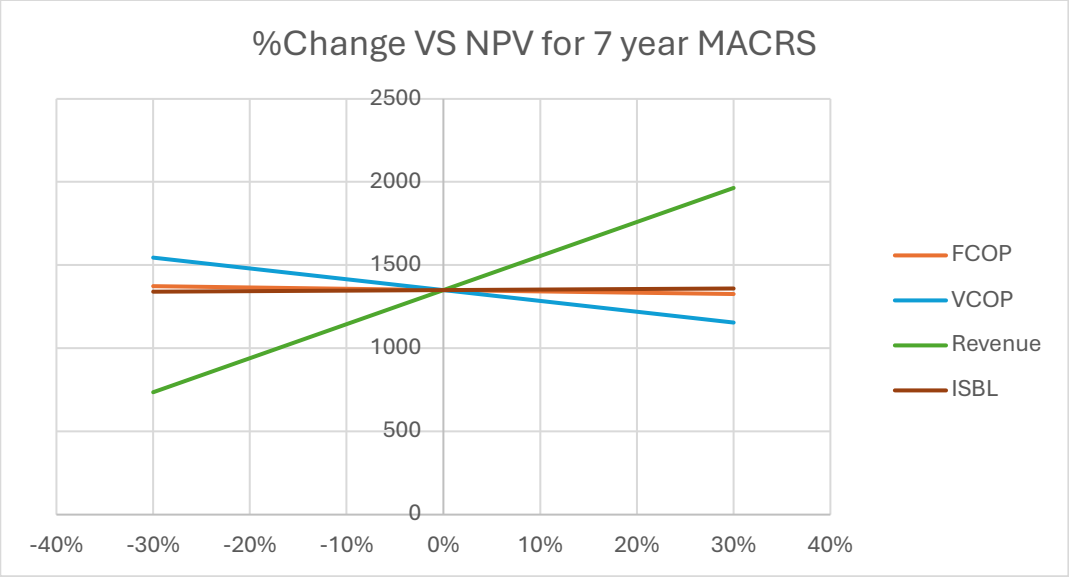


Figure 6: Sensitivity Analysis

Percentage	FCOP	VCOP	Revenue	ISBL
-30%	1373.84	1545.45	735.88	1340.61
-20%	1365.97	1480.37	940.67	1343.81
-10%	1358.09	1415.29	1145.44	1347.01
0%	1350.22	1350.21	1350.22	1350.22
10%	1342.34	1285.13	1554.99	1353.42
20%	1334.46	1220.05	1759.76	1356.62
30%	1326.58	1154.97	1964.54	1359.82

The sensitivity analysis shown in Figure 6 evaluates how uncertainties in key economic parameters affect the Net present value (NPV) of the process using a 7-year MACRS depreciation schedule. A variation of -30% to +30% was applied to the Fixed cost production (FCOP), Variable Cost of production (VCOP), Revenue, and the inside Battery limits (ISBL) cost to determine which parameter has the greatest influence on the plant's profits.

From the results, Revenue has the strongest impact on NPV. As Revenue increases, the NPV rises significantly, while a decrease in revenue caused a substantial drop in profitability. This indicates that the economic success of the plant is dependent on maintaining a strong product that sells Aswell with having a stable market demand. The steep sloe of the Revenue line demonstrates that even moderate changes in selling price or production output can greatly affect the financial performance of the project.

In contrast, FCOP, VCOP, and ISBL show relatively small changes in NPV across the same percentage range. The FCOP and the ISBL have very small slope changes, indicating that fluctuations in fixed operating costs and capital investment have only a minor effect on the overall profitability of the process.

VCOP has a slightly larger influence than the FCOP and ISBL, as a increase in raw material and utilities would reduce the NPV, but the effect is still considerably smaller than that of revenue change.

Overall the sensitivity analysis suggests that the project is economically robust because the NPV remains positive even when major parameters vary within the 30 percent window. The overall process is relatively insensitive to the uncertainties in operating and capital costs, while revenue remains the critical economic driver. Because Revenue is the main economic driver for the process maintain consistent production capacity, product quality and a market is the most important facts ensuring long term profitability and investment successes.

Appendix

Design Basis

The plant was designed to produce approximately 280 million pounds per year of renewable RP-1 rocket fuel. This production capacity was selected based on process throughput values reported in the design case developed by Pacific Northwest National Laboratory for biomass conversion through fast pyrolysis, hydrotreating, and hydrocracking.[6] The published process analyzed a commercial-scale biomass refinery intended to demonstrate the technical and economic feasibility of large-scale fuel production from lignocellulosic feedstocks.

This capacity was considered appropriate for the present design because the proposed process also relies on wood-derived biomass feedstocks and similar upgrading operations to produce hydrocarbon fuels. Using a production scale comparable to the published design allows equipment sizing, utility demands, and economic estimates to remain realistic for an industrial facility rather than a pilot-scale operation. In addition, a plant of this size provides sufficient production volume to support the growing demand for rocket-grade kerosene fuels in the aerospace industry while still remaining feasible with available biomass resources in California.[6 & 9]

Three primary criteria guided the selection of the plant location: feedstock availability, tax and policy incentives, and proximity to major aerospace and launch facilities. California meets all three criteria exceptionally well.

California is one of the largest lumber-producing states in the United States, providing a large and reliable supply of wood-based feedstocks such as wood chips, which are generated directly and indirectly from lumber-processing operations. This supports long-term feedstock availability and helps reduce transportation and raw material costs.[7]

In addition, California offers significant financial incentives for low-carbon and renewable fuel production, including tax incentives and carbon-reduction credit programs that improve overall project profitability.[11]

Finally, California is home to major aerospace and launch organizations, including SpaceX, NASA, and Northrop Grumman, which operate launch facilities and research centers within the state. Proximity to these organizations supports easier fuel distribution, opportunities for industrial partnerships, and long-term demand for renewable RP-1 fuel.

Block Flow Diagram (BFD)

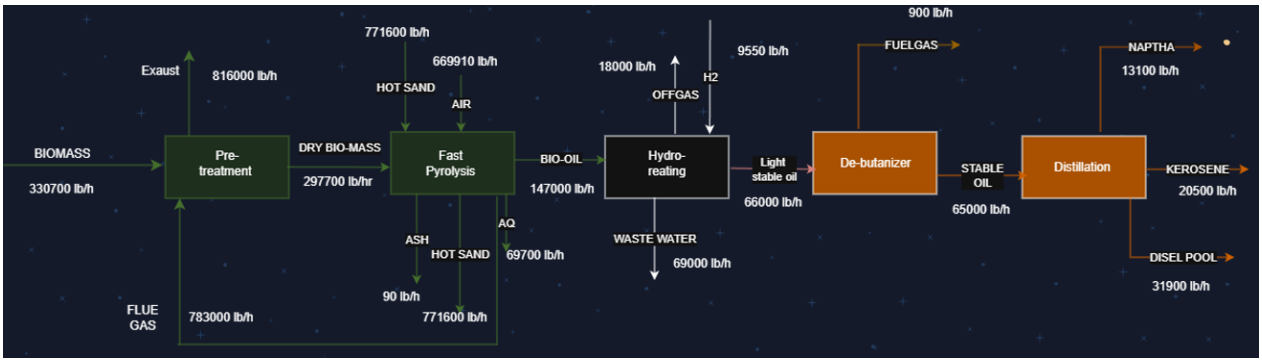


Figure 7: Block Flow Diagram

Preliminary Process Flow Diagram

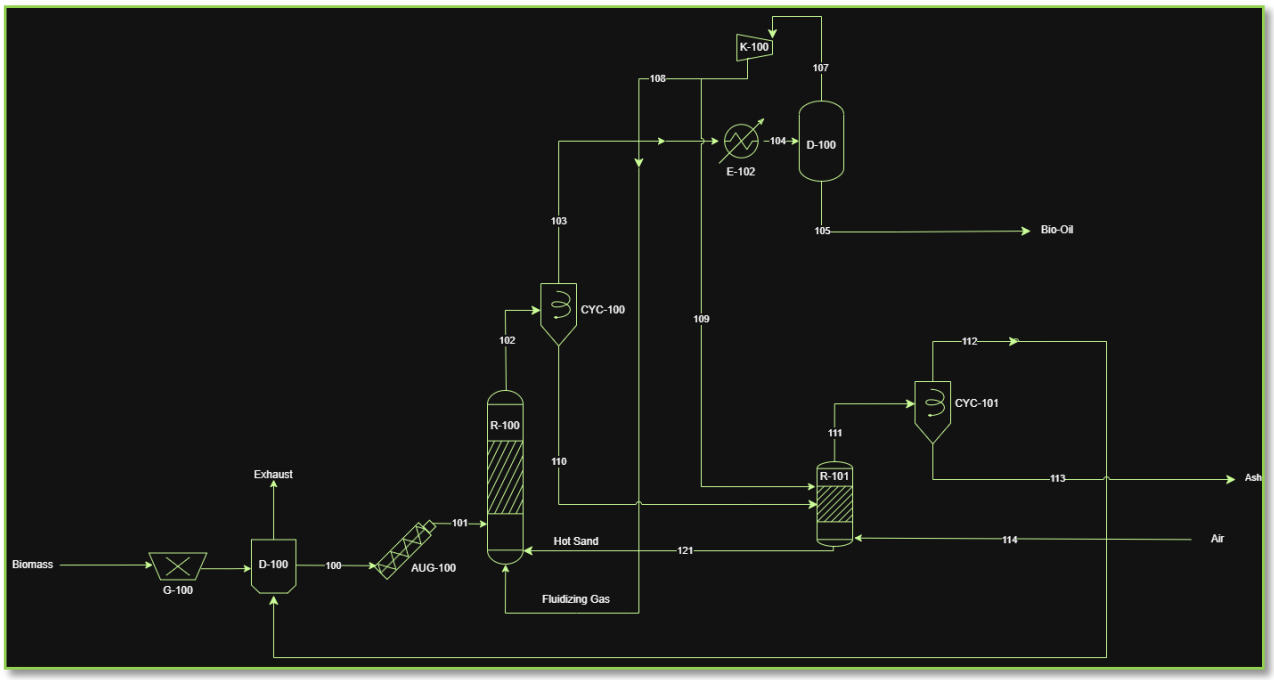


Figure 8: PFD of Fast Pyrolysis

Table 7: Fast Pyrolysis Mass Balance

	Biomass	100		110	121	AIR	ASH
TEMP(F)	77	250		896	2157	77	2157
PRESS(PSI)	33.8	16.5		29	16.5	14	16.5
WOOD(lb/hr)	330693	297624		-	-	-	-
SAND(lb/hr)	-	-		771617	771617	-	-
CHAR(lb/hr)	-	-		28869	-	-	86.6
N2(lb/hr)	-	-		-	-	513876	-
O2(lb/hr)	-	-		-	-	156033	-

	FLUEGAS	FLUIDGAS	102	111	112	109	Exhaust
Temp(F)	377	707	896	2157	2157	176	250
PRESS(PSI)	16.5	33.9	29.4	16.5	16.5	34	16.5
H2(lb/hr)	-	-	297.6	-	-	155	-
N2(lb/hr)	513905	-	-	513905	513905	-	462173
CO(lb/hr)	-	17605	34927	-	-	-	-
O2(lb/hr)	17118	-	-	17118	17118	-	17578
CO2(lb/hr)	18765	17835	35782	187625	187625	18692	264980
H2O(lb/hr)	63634	5478	82596	63634	63634	8629	69857
SO2(lb/hr)	659	1174	1174	659	659	613	1356
NO2(lb/hr)	145	278	278	145	145	145	214
C1-C2	-	1700	4528	-	-	2365	-
Oxygynates	-	33088	219401	-	-	-	-
SOLIDS(lb/hr)	-	-	800487	86	-	-	-

Streams	103	104	105	107
TEMP(F)	896	113	113	113
PRESS(PSI)	29.4	24.9	24.9	24.9
H2 lb/hr	297.6	297.6	-	297.6
N2 lb/hr	-	-	-	-
CO lb/hr	34927.6	34927.6	-	34927.6
O2 lb/hr	-	-	-	-
CO2 lb/hr	35782.2	35782.2	-	35782.2
SO2 lb/hr	1174.9	1174.9	-	1174.9

NO2 lb/hr	278.1	278.1	-	278.1
H2O lb/hr	82596.1	82596.1	66076.9	16519.2
C1-C2 lb/hr	4528	4528.1	-	4528.1
Oxygynates lb/hr	219401	219401	151349.9	68051.2
SOLIDS lb/hr	-	-	-	-

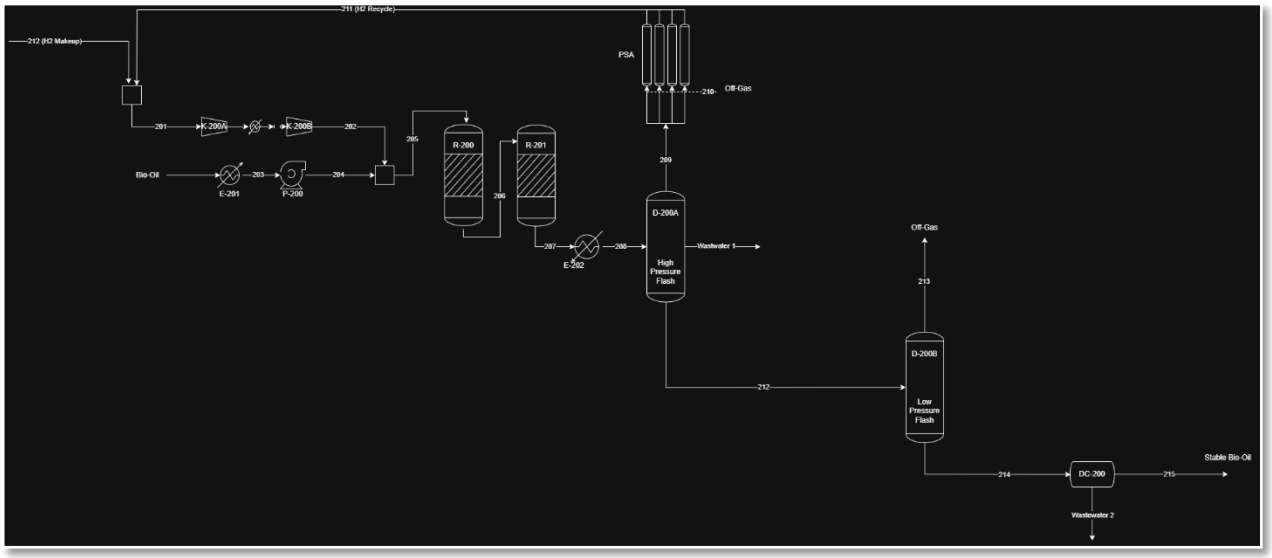


Figure 9: PFD for Bio-Oil Upgrading

Table 8: bio-Oil Upgrading

	BIO-OIL	201	202	203	204	205	206	207	208
TEMP(F)	285.7	99.7	248	212	216.9	185.4	185	149	149
PRESS(PSI)	24.9	300	30.9	13.7	1469.6	30.9	1450.4	1449.4	414.7
H2O lb/hr	101.7	-	-	101.7	101.7	101.7	70069.7	70069.7	615.2556
H2 lb/hr	-	9478.3	9478.3	-	-	9478.3	3740.2	3740.2	17.10703
CO2 lb/hr	-	-	-	-	-	-	8766.5	8766.5	550.9927
C1-C2 lb/hr	-	-	-	-	-	-	5218.1	5218	216.2984
C3-C4 lb/hr	-	-	-	-	-	-	3126.2	3126.2	820.7999

C5-C9 lb/hr	-	0.0001273	0.0001273	-	-	0.000127	22833.3	22833.3	22355.38
C10-C16 lb/hr	-	10.8	10.8	-	-	10.8	35242.3	35242.3	35193.39
C17 lb/hr	-	-	-	-	-	-	6245.4	6245.4	6245.394
Oxygenates lb/hr	147593.9	1.339E-09	1.339E-09	147593.9	147593.9	147593.9	0.00099	0.00099	0.000901

	209	210	211	212	213	214	215	WASTE Water	WASTE Water 2
Temp(F)	149	149	149	77	200	200	206.8	149	206.8
Press(PSI)	414.7	414.7	414.7	300	49.7	49.7	49.7	414.7	49.7
H2O lb/hr	391.4	391.4	-	-	76.6	538.6	10.02	69062.9	528.6
H2 lb/hr	3722.9	744.6	2978.3	6500	16.5	0.65063	0.650638	0.1725131	-
CO2 lb/hr	8207.5	8207.5	-	-	389.7	161.2	161.2	8.0	-
C1-C2 lb/hr	5001.1	5001.1	-	-	153.8	62.5	62.5	0.619	-
C3-C4 lb/hr	2305.4	2305.4	-	-	267.5	553.3	553.3	0.0404	-
C5-C9 lb/hr	457.9	457.9	0.00012735	-	117	22238.4	22238.4	19.901	-
C10—C16 lb/hr	48.9	38.1	10.8	-	15.4	35178	35178	3.4621E-05	-
C17 lb/hr	0.00514419	0.005144	-	-	0.002433037	6245.4	6245.4	4.8141E-13	-
Oxygenates	1.339E-09	-	1.339E-09	-	1.15085E-09	0.000901	0.000901	8.9665E-05	-

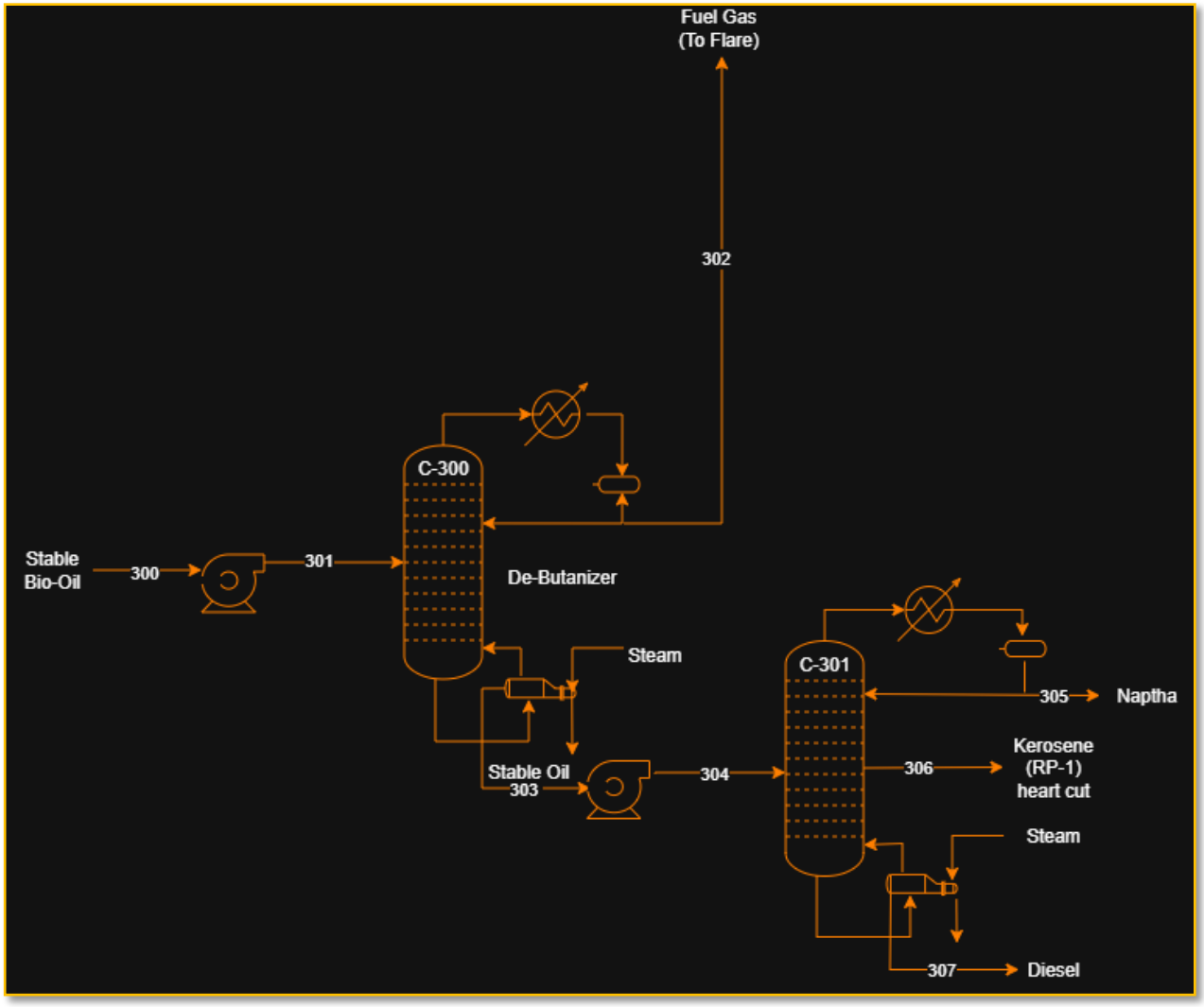


Figure 10: RP-1 Fractionation PFD

Table 9: RP-1 Fractionation Mass Balance

	300	301	302 (Fuel Gas)	303 (Stable Oil)	304	NAPTHA	RP-1	Diesel Pool
TEMP (F)	206.7	208.005	209.5	656.2	657.1	573.8	667.7	876.1
PRESS(PSI)	49.6	214.6	214.6	216.3	264.6	264.6	293.9	293.9
C1-C2 lb/hr	223.7	223.7	223.7	6.97324E-06	6.97324E-06	6.97322E-06	1.53469E-11	8.98E-21
C3-C4lb/hr	553.2	553.2	553.2	0.028	0.02	0.028	3.61941E-06	1.51E-13
C5-C9 lb/hr	22238.3	22238.3	1.16E+02	22122.1	22122.1	9816.9	11801.9	185.6
C10-C16 lb/hr	37079.0	37079.002	0.0019	37079.001	37079.0	3272.3	8696.5	27254.4

C17+ lb/hr	6245.3	6245.3	9.83426E-23	6245.3	6245.3	2.50592E-05	0.00019	4418.6
H2 lb/hr	0.6506	0.6506	0.6	9.66E-12	9.66E-12	9.66E-12	0	6.69E-31
H2O lb/hr	10.02	10.02	10.01	0.007	0.007	0.007	4.59E-06	1.06544E-12
Oxygenates lb/hr	0.0009	0.0009	6.40E-25	0.0009	0.0009	5.15E-10	3.33E-04	1.11E-08
Mass Flow lb/hr	66,350.4	66,350.4	903.8	65,446.5	65,446.5	13,089.3	20,498.5	31,858.7

Rough Material Balance

Table 10: General Material Balance around the BFD

Block	Pre-treatment	Fast Pyrolysis	Hydro-treating	De-butanizer	Distillation	Steam reforming
Stream	S1+S4=S2+S3	S3+S10=S4+S5	S6=S7+S11+S8	S11=S12	S12=S13+S14+S15	S8+S9=S10

At this point in the project, a general mass balance can be performed, meaning that the values in Figure 2 are not accurate, but rather rough estimates based on the literature. What is known for certain is that the average yield of RP-1 is around 50 percent, 25 percent Naphtha, and 25 percent Biodiesel, as well as the amount fed to the process. To make a safe assumption as to how much sustainable RP-1 can be produced from biomass, the assumption that 50 percent of Naphtha and diesel is made from biomass is instead RP-1.[4] About 70,000 lb/h of stable oil is to be separated into biodiesel and naphtha, applying the assumption that the theoretical amount of RP-1 that should yield is about 35,000lb/h of Kerosene(RP-1) and 17,500lb/h of naphtha and biodiesel.

Aspen Plus PFD

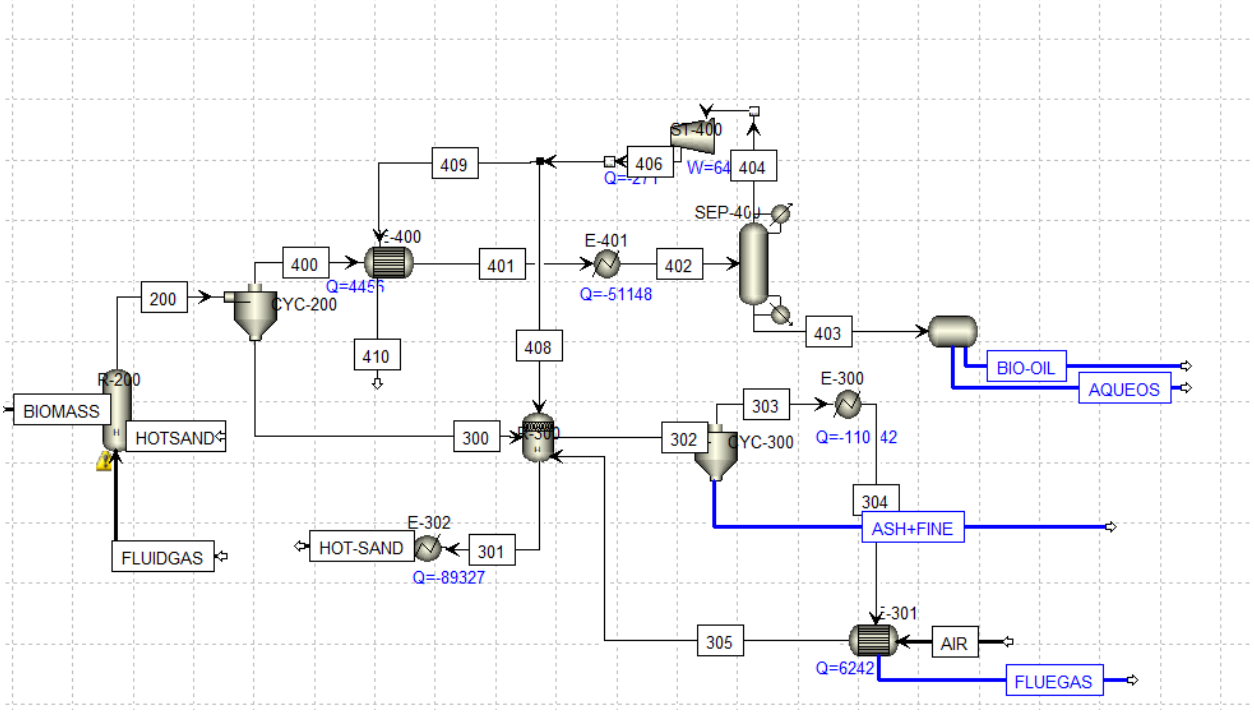


Figure 11: ASPEN PFD (FAST PYROLYSIS)

In this simulation, one of the design specifications was implemented in Aspen Plus code to accurately control the oxygen content of the combustor's effluent stream (S13) to 2 mol% O_2, matching the required amount for this assignment. The design specification was set to automatically change the molar flow rate of Air entering the combustor to match its desired amount directly through its manipulated variable ranges from 1000 to 6000 kmol/hr while its specified variable is the mole fraction of O_2 at 0.02 at S13 for Aspen to iteratively adjust its input Air flow to meet its desired amount directly for perfect execution and matching perfectly its desired amount of oxygen exiting the combustor at 2%. This amount is 2% because it is expected to behave similarly to industrial requirements for excess air requirements to guarantee complete burning of char while avoiding excess dilation from nitrogen addition to flue gas for industrial applications properly, while maintaining its heat correctly for repeated reheating for its primary role at the pyrolysis reactor to guarantee successful execution of its processes properly.

A further specification was added to the Aspen Plus input file to ensure adiabatic operation of the RYIELD reactor (R-100), specifying a reactor duty of 0 kW, which means no heat addition or removal is to occur. In this specification, the reactor duty for R-100 was specified as the specified variable and maintained at its target value of 0 kW, while the reactor temperature for E-103 was specified as the manipulated variable that Aspen should adjust to meet this condition. It should operate within reactor temperatures ranging from 100°C to 800°C, ensuring this is a realistic range for which Aspen can determine the correct inlet temperature to self-heat to its final condition solely by reaction heat, without requiring additional heat

input from external heating sources. This is also realistic for fast pyrolysis reactor designs, where biomass is heated solely by hot-sand circulation rather than requiring additional heat input components.

It is logical to use reactor temperature as this is directly dependent on whether or not reactor energy terms will reach closure without requiring heat interaction rather than being dependent on E-103 manipulated variable at all points during simulates reactor processes using Aspen plus software because inlet temperature to reactor-R100 is one factor that mainly affects whether or not energy terms reach closure at all points during simulates reactor processes using Aspen plus software because inlet temperature to reactor-R100 is one factor that mainly affects whether or not energy terms reach closure at all points during reactor processes.

The pyrolysis reactor was modeled using wood as the primary biomass feedstock. Because wood is a nonconventional solid, its physical-chemical behavior in Aspen Plus was defined using proximate analysis, ultimate analysis, and sulfur analysis. These values allow Aspen's nonconventional solid property models (HCOALGEN and DCOALIGT) to calculate sensible enthalpy, density, and heating values. The assignment indicated that the biomass contained 10% moisture, so the raw wood stream was dried prior to pyrolysis. This drying step removes only water, ensuring the reactor receives fully dried wood. The same proximate, ultimate, and sulfur analyses defined for wood were also used as the basis for downstream modeling of ash, with char receiving updated values reflecting its expected high fixed carbon, low volatile matter, and greatly reduced oxygen and hydrogen after pyrolysis.

The pyrolysis reactor (R-100) was modeled as an RYield reactor. The RYield block was used in this assignment since the detailed kinetics and reaction pathways were not provided, and the desired behavior was instead defined by experimentally measured pyrolysis reactor yields. In this model, each product stream from the assignment was assigned a mass-based yield for every 100 grams of dry wood entering the reactor. These yields included hydrogen, carbon monoxide, carbon dioxide, water, methane, ethylene, levoglucosan, guaiacol, acetone, acetic acid, cresol, furfural, and char. Sand was listed as an inert component so that it passes through the reactor unchanged. Aspen automatically normalizes these yields to maintain mass balance. Because the pyrolysis process is strongly endothermic, the internal heat balance of the reactor must be satisfied by the sand loop supplying heat.

To enforce adiabatic operation, a design specification (DS-ADIAB) was created. The specification forces the calculated reactor duty to be zero by manipulating the temperature of the incoming hot sand. The manipulated variable was the temperature of the heater block feeding hot sand into the reactor, and it was allowed to vary within physically reasonable bounds. During simulation, Aspen adjusts the sand temperature iteratively until the reactor requires no external heat or cooling, meaning that the sand supplies exactly the amount of heat needed to achieve the specified yield distribution. This approach mirrors the behavior of circulating fluidized-bed pyrolysis systems, where heat is transferred from a recirculating solid heat carrier rather than from an external furnace or internal heating elements.

Several assumptions were necessary to implement this model. Wood is treated as a uniform material with fixed proximate and ultimate analysis values, even though natural biomass varies significantly. The RYield reactor assumes that the product slate depends only on the specified yields; no reaction kinetics, residence-time effects, or secondary cracking reactions are considered. Gas-phase secondary reactions such as tar cracking or polymerization were neglected. Sand and ash were treated as chemically inert solids. The reactor was assumed to be perfectly mixed and adiabatic, with all heat provided solely by the incoming sand. Vapor-phase components were assumed to behave ideally under the selected property method. Sulfur

was assumed to remain primarily in ash or char unless explicitly represented in a vapor-phase product. Finally, the model assumes stable solids circulation and no agglomeration or defluidization behavior.

The char combustor was modeled using a two-stage RStoic configuration to represent the sequential oxidation and burnout of char produced in the pyrolysis reactor. The goal was to recover thermal energy from char combustion, generate a high-temperature fluidizing medium for the pyrolysis reactor, and produce a clean flue gas stream suitable for downstream heat recovery. The combustor was therefore modeled as a combination of R-200A (char oxidation stage) and R-200B (gas burnout and flue gas polishing stage), each operating at controlled pressure and near-zero duty to achieve a thermodynamically consistent representation of adiabatic combustion.

The first reactor, R-200A, receives the char stream generated in fast pyrolysis (DRYCHAR) and initiates the decomposition of the nonconventional char component into its elemental constituents. Because char is defined in Aspen using nonconventional component modeling (with PROXANAL, ULTANAL, and SULFANAL attributes), the combustion chemistry must be explicitly defined through stoichiometric reactions. In R-200A, we modeled the breakdown of char into elemental carbon (solid C), oxygen (O_2), nitrogen (N_2), hydrogen (H_2), sulfur (S), and inert ash. This step does not represent the physical chemistry of combustion itself, but rather converts the NC component into conventional components so Aspen Plus can compute stoichiometric oxidation in subsequent units. To achieve this, R-200A was configured with a fractional conversion of 1.0 on char, and with stoichiometric coefficients calculated using a custom calculator. Since all ash present in the char should remain solid throughout the combustion train, the PROXANAL and ULTANAL values for ASH were set to 100% ash and 0% of all other constituents to ensure the material balances were preserved.

Following char decomposition, the now-conventional elemental components flow to R-200B, which performs the true combustion calculations. R-200B was configured at 1.12 atm and set to “Generate Combustion Reactions” in Aspen Plus, which automatically constructs oxidation pathways for carbon, hydrogen, sulfur, and any remaining combustibles. Nitrous oxide (NO) was selected as the preferred nitrogen oxide product to reflect typical fluidized bed combustion conditions. In this reactor, all available oxygen from the air inlet is consumed stoichiometrically, producing CO_2, H_2O, SO_2, and trace amounts of NO. R-200B operates with a duty fixed at 0, making the reaction adiabatic. This is critical to representing the direct transfer of combustion heat to the circulating bed material without artificially imposing external heat addition or removal.

The thermal energy generated in R-200B is carried by the high-temperature flue gas and by the heated circulating solids stream (ASHSAND). These solids consist of the recycled bed material that loops back to the pyrolysis reactor to supply process heat. Cyclone (CYC-102) was used to mechanically separate coarse ash and fines from the flue gas, ensuring that unburned solids remain within the circulating bed loop while fine particulates exit via ASHFINES. The flue gas exiting R-200B and associated separators is sent to a series of heat exchangers (E-101, E-102, and E-100, depending on loops) to recover energy before being discharged as F-LUEGAS1 or F-LUEGAS2.

Air addition to R-200B was modeled explicitly via a dedicated air stream at 25°C and 1 atm. The amount of air supplied exceeds the stoichiometric requirement to ensure complete burnout of char and oxidation of volatile elements. Compressors and heat exchangers upstream of R-200B are included where needed to reflect required operating conditions but do not alter the fundamental combustion chemistry. The pressure was maintained at 1.12 atm throughout the combustor system to approximate slightly pressurized fluidized bed operation and to maintain consistency with the pyrolysis reactor.

Assumptions were required to construct a workable and thermodynamically consistent combustor model. It was assumed that char undergoes complete conversion in R-200A, with no residual volatile matter remaining after pyrolysis. The ULTANAL elemental distribution was assumed constant for all char fed to the combustor, even though real char composition varies with heating rate, feedstock, and residence time. All ash was treated as inert and nonreactive, with no catalytic or heat capacity effects beyond those captured by Aspen's NC property models. Combustion was assumed to be adiabatic, with no heat losses to the environment, and all reactions in R-200B were treated as kinetically unlimited and instantaneous. Air was supplied at standard composition, and NO formation was limited to thermodynamic product selection rather than detailed NOx kinetics. Finally, bed material inventory was assumed constant, and attrition losses were represented only through ASHFINES removal.

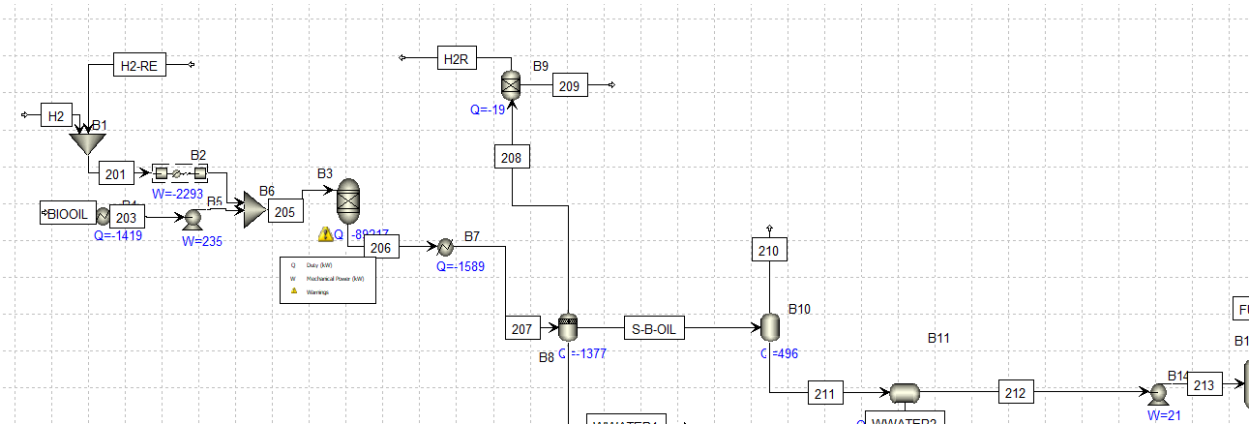


Figure 12: ASPEN PFD UPGRADING BIO-OIL

The filtered bio-oil product from the pyrolysis unit is pumped to high pressure, then combined with compressed hydrogen and preheated with reactor effluent. Two catalytic reaction stages are used. The first-stage catalytic reactor serves to stabilize the bio-oil through mild hydrotreatment over a cobalt-molybdenum (CoMo) hydrotreating catalyst. The product oil is further processed in the second-stage hydrotreater. The second-stage hydrotreater operates at a higher temperature and a lower space velocity than the first-stage hydrotreater. The CoMo catalyst is also used in this reactor. Reactor effluent and a fired heater provide feed preheat for the second stage. The second-stage product is separated into product oil, wastewater, and off gas streams.

The off-gas from the hydrotreaters is sent to a Pressure Swing Adsorption (PSA) system for recovery of the hydrogen gas. The recovered hydrogen is recycled back to the reactors. The low-pressure PSA tail gas stream, which is rich in byproduct light hydrocarbons, is sent to the steam reformer for hydrogen production. A small portion of this off-gas is used in the fired heater to preheat the feed to the second-stage hydrotreater.

Anaerobic digestion can be used to treat wastewater with an organic content of less than 2%.

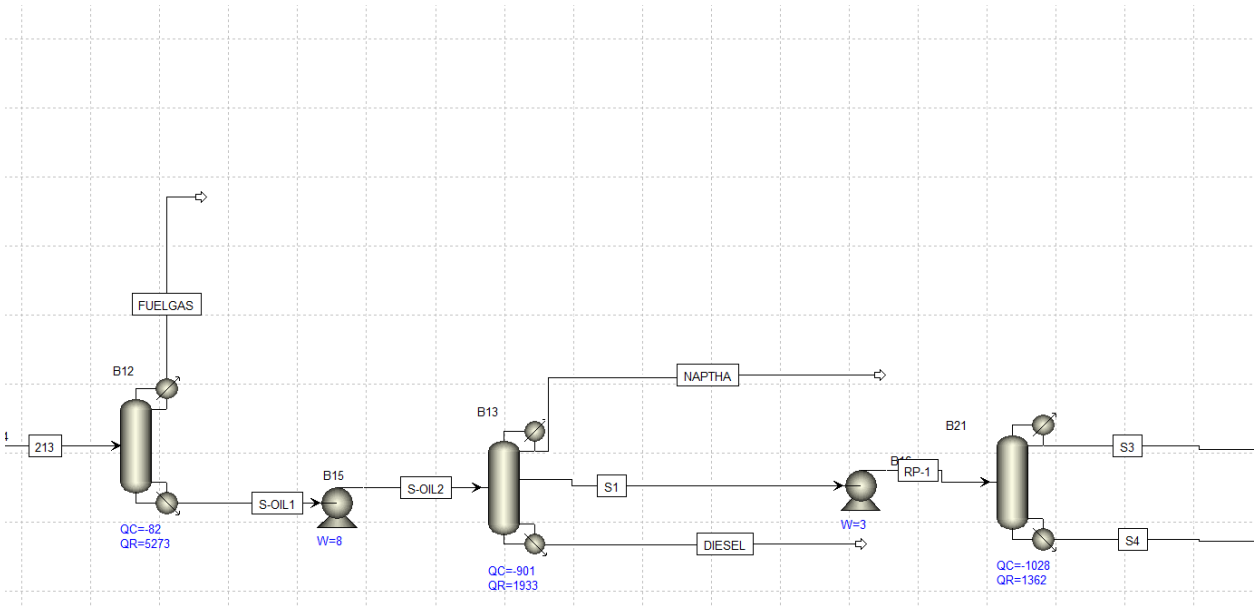


Figure 13: ASPEN PFD FRACTIONATION

This PFD shows the flow diagram for the hydrocracking and product separations areas. The hydrotreated oil is stabilized by removing the butane and lighter components in a debutanizer column. The stable oil stream is then separated into light and heavy fractions. The desired product, RP-1, which is the carbon range C9-C16, is extracted from a hard cut down in the middle. Side products such as Diesel (C12-C20) and Naphtha (C5-C7) are also obtained from the tops and bottoms.

MASS BALANCE (Fast Pyrolysis)

Table 11: Mass Balance Fast Pyrolysis

Column1	Column2	BIOMASS (100)	HOTSAND (123)	FLUIDGAS (113)
Temperature	F	77	1129.9	707
Pressure	psi	33.8	33.8	33.94
H2 lb/hr	lb/hr	0	0	0
N2 lb/hr	lb/hr	0	0	0
CO lb/hr	lb/hr	0	0	17605
O2 lb/hr	lb/hr	0	0	0
CH4 ib/hr	lb/hr	0	0	847
C2H4 lb/hr	lb/hr	0	0	853.7
CO2 lb/hr	lb/hr	0	0	17835.5
SO2 lb/hr	lb/hr	0	0	1174.9
NO2 lb/hr	lb/hr	0	0	278.1
C3H6O-1 lb/hr	lb/hr	0	0	33088.4
H2O lb/hr	lb/hr	0	0	5478.4

C2H4O2-1 lb/hr	lb/hr	0	0	0
C5H4O2 lb/hr	lb/hr	0	0	0
C7H8O-4 lb/hr	lb/hr	0	0	0
C7H8O2-E1 lb/hr	lb/hr	0	0	0
C6H10O5-N1 lb/hr	lb/hr	0	0	0
SAND lb/hr	lb/hr	0	771617.9	0
ASH lb/hr	lb/hr	0	0	0
WOOD lb/hr	lb/hr	330693.4	0	0
CHAR lb/hr	lb/hr	0	0	0

Column1	Column2	200	300	301	302
Temperature	F	896	896	2157.1	2157.1
Pressure	psi	29.4	29.4	16.5	16.4
H2 lb/hr	lb/hr	297.6	0	0	0
N2 lb/hr	lb/hr	0	0	0	5
CO lb/hr	lb/hr	34927.6	0	0	0
O2 lb/hr	lb/hr	0	0	0	17118.9
CH4 ib/hr	lb/hr	2424.4	0	0	0
C2H4 lb/hr	lb/hr	2103.7	0	0	0
CO2 lb/hr	lb/hr	35782.2	0	0	187625.8
SO2 lb/hr	lb/hr	1174.9	0	0	659.9
NO2 lb/hr	lb/hr	278.1	0	0	145.3
C3H6O-1 lb/hr	lb/hr	70886.6	0	0	0
H2O lb/hr	lb/hr	82596	0	0	63634.6
C2H4O2-1 lb/hr	lb/hr	33036.2	0	0	0
C5H4O2 lb/hr	lb/hr	8631.1	0	0	0
C7H8O-4 lb/hr	lb/hr	11904.9	0	0	0
C7H8O2-E1 lb/hr	lb/hr	81846.6	0	0	0
C6H10O5-N1 lb/hr	lb/hr	13095.5	0	0	0
SAND lb/hr	lb/hr	771617.9	771617.9	771617.9	0
ASH lb/hr	lb/hr	0	0	0	86.6
WOOD lb/hr	lb/hr	0	0	0	0
CHAR lb/hr	lb/hr	28869.5	‘28869.5	0	0

Column1	Column2	303	304	305	400
Temperature	F	2157.6	482	212	896
Pressure	psi	16.4	16.4	14.6	29.3
H2 lb/hr	lb/hr	0	0	0	297.6
N2 lb/hr	lb/hr	513905.2	513905.2	513876.3	0

CO lb/hr	lb/hr	0	0	0	34927.5
O2 lb/hr	lb/hr	17118.9	17118.9	156033.3	0
CH4 ib/hr	lb/hr	0	0	0	2424.4
C2H4 lb/hr	lb/hr	0	0	0	2103.6
CO2 lb/hr	lb/hr	187625.7	187625.7	0	35782.1
SO2 lb/hr	lb/hr	659.8	659.8	0	1174.8
NO2 lb/hr	lb/hr	145.2	145.2	0	278.1
C3H6O-1 lb/hr	lb/hr	0	0	0	70886.6
H2O lb/hr	lb/hr	63634.6	63634.6	0	82596.1
C2H4O2-1 lb/hr	lb/hr	0	0	0	33036.3
C5H4O2 lb/hr	lb/hr	0	0	0	8631.1
C7H8O-4 lb/hr	lb/hr	0	0	0	11904.9
C7H8O2-E1 lb/hr	lb/hr	0	0	0	81846.6
C6H10O5-N1 lb/hr	lb/hr	0	0	0	13095.5
SAND lb/hr	lb/hr	0	0	0	0
ASH lb/hr	lb/hr	0	0	0	0
WOOD lb/hr	lb/hr	0	0	0	0
CHAR lb/hr	lb/hr	0	0	0	0

Column1	Column2	401	402	403	404
Temperature	F	807.8	113	113	113
Pressure	psi	29.4	24.9	24.9	24.9
H2 lb/hr	lb/hr	297.6	297.6	0	297.6
N2 lb/hr	lb/hr	0	0	0	0
CO lb/hr	lb/hr	34927.6	34927.6	0	34927.6
O2 lb/hr	lb/hr	0	0	0	0
CH4 ib/hr	lb/hr	2424.4	2424.4	0	2424.4
C2H4 lb/hr	lb/hr	2103.6	2103.7	0	2103.7
CO2 lb/hr	lb/hr	35782.1	35782.1	0	35782.2
SO2 lb/hr	lb/hr	1174.8	1174.8	0	1174.9
NO2 lb/hr	lb/hr	278.1	278.1	0	278.1
C3H6O-1 lb/hr	lb/hr	70886.6	70886.6	2835.5	68051.2
H2O lb/hr	lb/hr	82596.1	82596.1	66076.9	16519.2
C2H4O2-1 lb/hr	lb/hr	33036.3	33036.7	33036.7	0
C5H4O2 lb/hr	lb/hr	8631.1	8631.1	8631.1	0
C7H8O-4 lb/hr	lb/hr	11904.9	11904.9	11904.9	0
C7H8O2-E1 lb/hr	lb/hr	81846.6	81846.6	81846.6	0
C6H10O5-N1 lb/hr	lb/hr	13095.5	13095.5	13095.5	0
SAND lb/hr	lb/hr	0	0	0	0
ASH lb/hr	lb/hr	0	0	0	0

WOOD lb/hr	lb/hr	0	0	0	0
CHAR lb/hr	lb/hr	0	0	0	0

Column1	Column2	409	410	BIO-OIL	AQUEOS
Temperature	F	176	707	191.6	191.6
Pressure	psi	33.9	33.9	24.9	24.9
H2 lb/hr	lb/hr	142.1	142.1	0	0
N2 lb/hr	lb/hr	0	0	0	0
CO lb/hr	lb/hr	16681.7	16681.6	0	0
O2 lb/hr	lb/hr	0	0	0	0
CH4 ib/hr	lb/hr	1157.9	1157.93	0	0
C2H4 lb/hr	lb/hr	1004.7	1004.73	0	0
CO2 lb/hr	lb/hr	17089.8	17089.8	0	0
SO2 lb/hr	lb/hr	561.1	561.1	0	0
NO2 lb/hr	lb/hr	132.8	132.8	0	0
C3H6O-1 lb/hr	lb/hr	32501.7	32501.7	2834.2	1.17
H2O lb/hr	lb/hr	7889.7	7889.7	67.6	66009.2
C2H4O2-1 lb/hr	lb/hr	0	0	32789.2	246.9
C5H4O2 lb/hr	lb/hr	0	0	8598.7	32.4
C7H8O-4 lb/hr	lb/hr	0	0	11902.9	2.00
C7H8O2-E1 lb/hr	lb/hr	0	0	81792.3	54.2
C6H10O5-N1 lb/hr	lb/hr	0	0	13004.8	90.6
SAND lb/hr	lb/hr	0	0	0	0
ASH lb/hr	lb/hr	0	0	0	0
WOOD lb/hr	lb/hr	0	0	0	0
CHAR lb/hr	lb/hr	0	0	0	0

MASS BALANCE (BIO-OIL UPGRADE to RP-1)

Table 12: Mass Balance Bio-Oil Upgrading

Column1	Column2	H2	BIOIL	H2-RE
Temperature	F	77	191.6	149
Pressure	psi	300	24.9	414.6
H2	lb/hr	6500	0	3041.9
CH4	lb/hr	0	0	0
C2H4	lb/hr	0	0	0
CO2	lb/hr	0	0	0
ACETONE	lb/hr	0	2834.2	0
H2O	lb/hr	0	67.6	0
AACID	lb/hr	0	32789.2	0

FURFURAL	lb/hr	0	8598.7	0
MCRESOL	lb/hr	0	11902.9	0
GUAIACOL	lb/hr	0	81792.3	0
LVG	lb/hr	0	13004.8	1.36761E-09
C2H6	lb/hr	0	0	0
C10H22	lb/hr	0	0	0
2-PHE-01	lb/hr	0	0	0
N-HEP-01	lb/hr	0	0	0
P-XYL-01	lb/hr	0	0	0
1:3:5-01	lb/hr	0	0	0
N-PRO-01	lb/hr	0	0	0.00013
PHENA-01	lb/hr	0	0	0
CHRYS-01	lb/hr	0	0	0
DIAMA-01	lb/hr	0	0	0
DIPHE-01	lb/hr	0	0	0
DECAH-01	lb/hr	0	0	10.9
N-BUT-03	lb/hr	0	0	0
N-PRO-02	lb/hr	0	0	0
1:3:5-02	lb/hr	0	0	0
PROPA-01	lb/hr	0	0	0
ISOBU-01	lb/hr	0	0	0
1:1-D-01	lb/hr	0	0	0
N-TRI-01	lb/hr	0	0	0
BICYC-01	lb/hr	0	0	0
TETRA-01	lb/hr	0	0	0

Column1	Column2	201	202	203	204
Temperature	F	100	248	212	216.8
Pressure	psi	300	30.89	13.69	1469.5
H2	lb/hr	9541.98	9541.98	0	0
CH4	lb/hr	0	0	0	0
C2H4	lb/hr	0	0	0	0
CO2	lb/hr	0	0	0	0
ACETONE	lb/hr	0	0	2834.29	2834.29
H2O	lb/hr	0	0	67.66	67.6
AACID	lb/hr	0	0	32789.2	32789.2
FURFURAL	lb/hr	0	0	8598.7	8598.7
MCRESOL	lb/hr	0	0	11902.9	11902.9
GUAIACOL	lb/hr	0	0	81792.3	81792.3
LVG	lb/hr	1.36761E-09	1.36761E-09	13004.8	13004.8

C2H6	lb/hr	0	0	0	0
C10H22	lb/hr	0	0	0	0
2-PHE-01	lb/hr	0	0	0	0
N-HEP-01	lb/hr	0	0	0	0
P-XYL-01	lb/hr	0	0	0	0
1:3:5-01	lb/hr	0	0	0	0
N-PRO-01	lb/hr	0.000130	0.000130	0	0
PHENA-01	lb/hr	0	0	0	0
CHRYS-01	lb/hr	0	0	0	0
DIAMA-01	lb/hr	0	0	0	0
DIPHE-01	lb/hr	0	0	0	0
DECAH-01	lb/hr	1.10E+01	10.98	0	0
N-BUT-03	lb/hr	0	0	0	0
N-PRO-02	lb/hr	0	0	0	0
1:3:5-02	lb/hr	0	0	0	0
PROPA-01	lb/hr	0	0	0	0
ISOBU-01	lb/hr	0	0	0	0
1:1-D-01	lb/hr	0	0	0	0
N-TRI-01	lb/hr	0	0	0	0
BICYC-01	lb/hr	0	0	0	0
TETRA-01	lb/hr	0	0	0	0

Column1	Column2	205	206	207	208
Temperature	F	185.3	185	149	149
Pressure	psi	30.8	1450.3	1449.3	414.69
H2	lb/hr	9541.9	3820.1	3820.1	3802.4
CH4	lb/hr	0	3332.8	3332.87	3271.8
C2H4	lb/hr	0	0	0	0
CO2	lb/hr	0	8953.8	8953.8	8382.8
ACETONE	lb/hr	2834.2	0	0	0
H2O	lb/hr	67.6	71566.8	71566.8	399.74
AACID	lb/hr	32789.2	0	0	0
FURFURAL	lb/hr	8598.7	0	0	0
MCRESOL	lb/hr	11902.9	0	0	0
GUAIACOL	lb/hr	81792.3	0	0	0
LVG	lb/hr	13004.8	0.0010	0.00101	1.36761E-09
C2H6	lb/hr	0	1996.6	1996.6	1836.12
C10H22	lb/hr	0	1984.4	1984.4	33.5
2-PHE-01	lb/hr	0	7958.9	7958.9	11.5
N-HEP-01	lb/hr	0	3258.8	3258.8	272.6
P-XYL-01	lb/hr	0	1729.2	1729.2	34.50

1:3:5-01	lb/hr	0	638.8	638.8	4.78
N-PRO-01	lb/hr	0.00013	0.0130	0.013	0.00013
PHENA-01	lb/hr	0	6056.6	6056.6	0.0782
CHRYS-01	lb/hr	0	6088.9	6088.9	0.00030
DIAMA-01	lb/hr	0	8590.9	8590.9	12.53
DIPHE-01	lb/hr	0	4360.8	4360.8	1.31
DECAH-01	lb/hr	1.10E+01	3170.0	3170.08	10.98
N-BUT-03	lb/hr	0	232.9	232.9	1.71
N-PRO-02	lb/hr	0	9735.3	9735.3	144.21
1:3:5-02	lb/hr	0	3170.0	3170.	3.58
PROPA-01	lb/hr	0	1696.2	1696.2	1359.34
ISOBU-01	lb/hr	0	1496.7	1496.7	995.3
1:1-D-01	lb/hr	0	1585.0	1585.0	13.16
N-TRI-01	lb/hr	0	8590.9	8590.9	6.39
BICYC-01	lb/hr	0	237.7	237.7	0.148
TETRA-01	lb/hr	0	289.9	289.9	0.004

Column1	Column2	209	S-B-OIL	WWATER1	210
Temperature	F	149	149	149	200
Pressure	psi	414.69	414.69	414.69	49.69
H2	lb/hr	760.49	17.47	0.176	16.80
CH4	lb/hr	3271.88	60.64	0.358	53.50
C2H4	lb/hr	0	0	0	0
CO2	lb/hr	8382.88	562.76	8.17	398.07
ACETONE	lb/hr	0	0	0	0
H2O	lb/hr	399.74	628.40	70538.5	78.27
AACID	lb/hr	0	0	0	0
FURFURAL	lb/hr	0	0	0	0
MCRESOL	lb/hr	0	0	0	0
GUAIACOL	lb/hr	0	0	0	0
LVG	lb/hr	0	0.00091	9.15805E-05	1.17544E-09
C2H6	lb/hr	1836.12	160.2	0.27	103.57
C10H22	lb/hr	33.55	1950.9	1.35443E-11	9.29
2-PHE-01	lb/hr	11.53	7927.03	20.32	4.79
N-HEP-01	lb/hr	272.6	2986.18	3.89652E-07	64.76
P-XYL-01	lb/hr	34.5	1694.6	0.000180	9.792
1:3:5-01	lb/hr	4.789	634.0	1.69828E-06	1.517187772
N-PRO-01	lb/hr	0	0.012	4.66E-11	3.90881E-05
PHENA-01	lb/hr	0.078	6056.5	2.00292E-06	0.042
CHRYS-01	lb/hr	0.00030	6088.	4.917E-13	0.00
DIAMA-01	lb/hr	12.531	8578.4	8.23947E-16	3.385
DIPHE-01	lb/hr	1.3172	4359.5	3.04595E-05	0.553

DECAH-01	lb/hr	0.00E+00	3159.0	1.6749E-06	3.450
N-BUT-03	lb/hr	1.710	231.2	2.28112E-11	0.478
N-PRO-02	lb/hr	144.21	9591.0	9.7275E-08	38.66
1:3:5-02	lb/hr	3.587	3166.4	1.20631E-11	1.3373
PROPA-01	lb/hr	1359.3	336.9	0.039	138.73
ISOBU-01	lb/hr	995.31	501.4	0.0019	134.48
1:1-D-01	lb/hr	13.16	1571.8	1.22296E-06	3.999
N-TRI-01	lb/hr	6.39	8584.5	1.92331E-19	2.419
BICYC-01	lb/hr	0.148	237.6	2.25016E-11	0.0503
TETRA-01	lb/hr	0.0049	289.93	7.21037E-47	0.00221

Column1	Column2	211	WWATER 2	212	213
Temperature	F	200	206.7670766	206.76	207.99
Pressure	psi	49.6	49.68	49.69	214.69
H2	lb/hr	0.66	0	0.66	0.66
CH4	lb/hr	7.13	0	7.13	7.13
C2H4	lb/hr	0	0	0	0
CO2	lb/hr	164.69	0	164.69	164.6
ACETONE	lb/hr	0	0	0	0
H2O	lb/hr	550.12	539.88	10.23	10.2
AACID	lb/hr	0	0	0	0
FURFURAL	lb/hr	0	0	0	0
MCRESOL	lb/hr	0	0	0	0
GUAIACOL	lb/hr	0	0	0	0
LVG	lb/hr	0.00091	0	0.00091	0.00091
C2H6	lb/hr	56.709	0	56.71	56.70
C10H22	lb/hr	1941.61	0	1941.61	1941.61
2-PHE-01	lb/hr	7922.2	0	7922.24	7922.24
N-HEP-01	lb/hr	2921.42	0	2921.42	2921.42
P-XYL-01	lb/hr	1684.90	0	1684.90	1684.90
1:3:5-01	lb/hr	632.49	0	632.49	632.4
N-PRO-01	lb/hr	0.012	0	0.01286	0.0128
PHENA-01	lb/hr	6056.5	0	6056.51	6056.5
CHRYS-01	lb/hr	6088.9	0	6088.90	6088.9
DIAMA-01	lb/hr	8575.0	0	8575.04	8575.04
DIPHE-01	lb/hr	4358.94	0	4358.94	4358.94
DECAH-01	lb/hr	3.16E+03	0	3155.64	3155.64
N-BUT-03	lb/hr	230.758	0	230.758	230.75
N-PRO-02	lb/hr	9552.42	0	9552.42	9552.4
1:3:5-02	lb/hr	3165.15	0	3165.150	3165.15

PROPA-01	lb/hr	198.176	0	198.176	198.176
ISOBU-01	lb/hr	366.93	0	366.93	366.936
1:1-D-01	lb/hr	1567.87	0	1567.87	1567.87
N-TRI-01	lb/hr	8582.14	0	8582.14	8582.140
BICYC-01	lb/hr	237.56	0	237.56	237.565
TETRA-01	lb/hr	289.93	0	289.93	289.932

Column1	Column2	FLUEGAS	S-OIL1	S-OIL2	DIESEL
Temperature	F	201.79	656.09	656.98	885.10
Pressure	psi	214.69	216.39	264.69	268.0
H2	lb/hr	0.66	9.95209E-12	9.95209E-12	5.95024E-31
CH4	lb/hr	7.13	3.11253E-09	3.11253E-09	6.78771E-27
C2H4	lb/hr	0	0	0	0
CO2	lb/hr	164.69	1.61972E-06	1.61972E-06	1.27643E-22
ACETONE	lb/hr	0	0	0	0
H2O	lb/hr	10.22	0.0075	0.0075	1.18785E-12
AACID	lb/hr	0	0	0	0
FURFURAL	lb/hr	0	0	0	0
MCRESOL	lb/hr	0	0	0	0
GUAIACOL	lb/hr	0	0	0	0
LVG	lb/hr	3.30434E-25	0.00091	0.00091	0.0005
C2H6	lb/hr	56.70	5.64116E-06	5.64116E-06	8.95106E-21
C10H22	lb/hr	0.00126	1941.61	1941.6	3.36
2-PHE-01	lb/hr	1.7433E-05	7922.24	7922.2	245.54
N-HEP-01	lb/hr	99.35	2822.07	2822.07	0.0062
P-XYL-01	lb/hr	0.0312	1684.87	1684.87	1.35
1:3:5-01	lb/hr	8.97189E-05	632.49	632.49	4.27
N-PRO-01	lb/hr	4.67906E-09	0.0128	0.0128	8.85919E-05
PHENA-01	lb/hr	6.93952E-17	6056.51	6056.51	3894.5
CHRYS-01	lb/hr	1.69731E-28	6088.90	6088.90	4188.1
DIAMA-01	lb/hr	2.85876E-10	8575.04	8575.04	4955.6
DIPHE-01	lb/hr	2.00927E-10	4358.94	4358.94	2224.00
DECAH-01	lb/hr	2.66E-06	3155.64	3155.64	599.8
N-BUT-03	lb/hr	1.16943E-06	230.75	230.75	23.7
N-PRO-02	lb/hr	0.0032	9552.42	9552.42	152.8
1:3:5-02	lb/hr	1.34191E-08	3165.15	3165.15	698.5

PROPA-01	lb/hr	198.17	0.00094	0.00094	1.72158E-16
ISOBU-01	lb/hr	366.9074144	0.02941	0.0294	1.62269E-13
1:1-D-01	lb/hr	0.000180466	1567.87	1567.87	19.5
N-TRI-01	lb/hr	4.7522E-10	8582.14	8582.14	2523.4
BICYC-01	lb/hr	2.81807E-11	237.56	237.56	110.03
TETRA-01	lb/hr	5.08837E-23	289.93	289.93	184.163

Column1	Column2	NAPTHA	RP1	RP-1
Temperature	F	573.4	705.02	705.02
Pressure	psi	264.6	293.91	293.91
H2	lb/hr	9.95209E-12	2.09284E-19	2.09284E-19
CH4	lb/hr	3.11253E-09	3.80721E-16	3.80721E-16
C2H4	lb/hr	0	0	0
CO2	lb/hr	1.61972E-06	1.14523E-12	1.14523E-12
ACETONE	lb/hr	0	0	0
H2O	lb/hr	0.0075	4.75578E-06	4.75578E-06
AACID	lb/hr	0	0	0
FURFURAL	lb/hr	0	0	0
MCRESOL	lb/hr	0	0	0
GUAIACOL	lb/hr	0	0	0
LVG	lb/hr	5.13834E-10	0.00033	0.00033
C2H6	lb/hr	5.64115E-06	1.5567E-11	1.5567E-11
C10H22	lb/hr	1106.5	831.68	831.68
2-PHE-01	lb/hr	2312.1	5364.56	5364.56
N-HEP-01	lb/hr	2693.	129.002	129.002
P-XYL-01	lb/hr	1078.7	604.81	604.81
1:3:5-01	lb/hr	277.2	350.96	350.96
N-PRO-01	lb/hr	0.0057	0.007065	0.00706
PHENA-01	lb/hr	0.0023	2161.99	2161.9
CHRYS-01	lb/hr	1.59702E-10	1900.778	1900.77
DIAMA-01	lb/hr	5.5	3613.82	3613.82
DIPHE-01	lb/hr	39.6	2095.29	2095.29
DECAH-01	lb/hr	5.09E+02	2047.16	2047.16
N-BUT-03	lb/hr	51.7	155.31	155.31
N-PRO-02	lb/hr	3677.6	5721.97	5721.97
1:3:5-02	lb/hr	385.8	2080.65	2080.65
PROPA-01	lb/hr	0.00094	2.41866E-8	2.41866E-08

ISOBU-01	lb/hr	0.029	3.69862E-06	3.69862E-06
1:1-D-01	lb/hr	614.92	933.37	933.37
N-TRI-01	lb/hr	616.07	5442.64	5442.6
BICYC-01	lb/hr	4.97	122.54	122.6
TETRA-01	lb/hr	2.50035E-05	105.76	105.76

Equipment List & Sizing

The equipment list labels and sizing were derived directly from Aspen Plus V14 software economic analysis function, focusing on the major components of the process design. Measurements were manually converted from the metric to imperial units. Material justifications were made based on the specific chemical and thermal requirements of each process stream.

For the heaters, Stainless Steel was selected to withstand high temperatures and acidity inherent in the process. Both Carbon Steel and Killed Carbon Steel were chosen for the various drums primarily for their low costs in neutralized environments. For the hydrotreater (R-200), 1 ¼ Chrome was selected as safeguard against High-Temperature Hydrogen Attack (HTHA) and sulfidation during the bio-oil upgrading phase. This is to prevent the vessel walls from becoming brittle and failing during high pressure operation.

Table 13: Fast Pyrolysis Equipment List

Fast Pyrolysis			
Label	Equipment	Materials of Construction	Sizing
D-100	Dryer	Carbon Steel	Tray Area = 200.21 sq ft
R-100	Pyrolysis Rise Reactor	316 SS	Diameter = 0.98 in Length = 20.01 ft
E-105	Heater	316 SS	Diameter = 0.98 in Length = 20.01 ft

E-106	Sheel-and-tube HEX	316 SS	Diameter = 0.98 in Length = 20.01 ft
E-107	Heater	316 SS	Diameter = 0.98 in Length = 20.01 ft
R-101	Combustor	316 SS	Volume= 38.2 gal Diameter = 12.01 in Height = 6.5 ft
D-100	Quench Absorber	Carbon Steel	Volume= 4,894.1 gal Diameter = 6.99 ft Height = 17.06 ft

Table 14: Bio-Oil Upgrading Equipment List

Bio-Oil Upgrading			
Label	Equipment	Materials of Construction	Sizing
R-200	Hydrotreater	1 ¼ Chrome	Volume = 2,022.55 gal Diameter = 4.49 ft Height = 17.06 ft
E-201	Heater	316 SS	Diameter = 0.98 in Length = 20.01 ft
E-202	Heater	316 SS	Diameter = 0.98 in Length = 20.01 ft
D-200A	Flash Drum	Killed Carbon Steel	Volume = 2,023.21 gal Diameter = 4.99 ft Height = 14.99 ft
D-200B	Flash Drum	Killed Carbon Steel	Volume = 2,023.21 gal Diameter = 4.99 ft Height = 14.99 ft
PSA	PSA	316 Stainless Steel	Volume = 634.52 gal Diameter = 2.99 ft Height = 12.01 ft

Table 15: Fractionation Equipment List

Fractionation			
Label	Equipment	Materials of Construction	Sizing
C-300	Distillation Column	Killed Carbon Steel	Diameter = 4 ft Height = 121.39 ft

C-301	Distillation Column	Killed Carbon Steel	Diameter = 4 ft Height = 121.39 ft

Hand Calculations

To begin the process of sizing the debutanizer distillation column, the Fenkse Equation is used to find N_{min}:

$$N_{min} = \frac{\log[\left(\frac{x_{LK,\,D}}{x_{HK,\,D}}\right)\left(\frac{x_{HK,\,B}}{x_{LK,\,B}}\right)]}{\log(\alpha_{LK,\,HK})}$$

Also, the geometric mean of the relative volatility is used:

$$\alpha_{LK,\,HK} = \sqrt{(\alpha_D)(\alpha_B)}$$

To start with, the relative volatility of the distillate is found. Antoine's equation is used to obtain the saturated pressures of the light and heavy key.

$$\log(P^{sat}) = A - \frac{B}{T + C}$$

A, B, and C are Antoine's coefficients, obtained from NIST. The relationship between the saturated pressures of the light and heavy key is as follows:

$$\alpha_D = \frac{P_{LK}^{sat}}{P_{HK}^{sat}}$$

For the light key, A = 4.3281, B = 1132.108, C = 0.918, and T = 366.3 K.

$$P_{LK}^{sat} = 17.64\ bar$$

For the heavy key, A = 4.01356, B = 1461.715, C = -65.16, and T = 619.7 K.

$$P_{HK}^{sat} = 0.145\ bar$$

From here:

$$\alpha_D = \frac{P_{LK}^{sat}}{P_{HK}^{sat}} = \frac{17.64}{0.145} = 121.45$$

Moving on the relative volatility of the bottoms. Since the temperature of the bottoms is above the critical temperature of the light key, Antoine's equation is not valid. Instead, Wilson's correlation is used.

$$\ln(k) = \ln\left(\frac{P_C}{P}\right) + 5.373(1+w)\left(1 - \frac{T_C}{T}\right)$$

$$\alpha_B = \frac{k_{LK}}{k_{HK}}$$

For the light key: Pc = 36.4 bar, w = 0.183, and Tc = 407.85 K.

$$k_{LK} = 17.54$$

For the heavy key, Pc = 28 bar, w = 0.258, and Tc = 639 K.

$$k_{HK} = 1.245$$

$$\alpha_B = \frac{k_{LK}}{k_{HK}} = \frac{17.54}{1.245} = 14.09$$

From here, all values required for N_{min} are known. Plugging everything into the Fenske equation results in:

$$N_{min} = 6.54 = 7\ stages$$

Next, N and N_{actual} are found. N is found using the 1st and 2nd Underwood Equations and the Gilliland Corelation:

$$F(1-q) = \sum_{i=1}^{C} \frac{\alpha_{i-ref} F z_{F,i}}{\alpha_{i-ref} - \theta}$$

$$V_{min} = \sum_{i=1}^{C} \frac{\alpha_{i-ref} (Dx_i)_D}{\alpha_{i-ref} - \theta}$$

$$\frac{V_{min}}{D} = R_{min} + 1$$

$$X = \frac{R - R_{min}}{R + 1}$$

$$Y = 1 - \exp\left[\frac{-1.33X}{1 - X}\right] = \frac{N - N_{min}}{N + 1}$$

Plugging everything in grants values of:

$$\theta = 39.87$$

$$R_{min} = 26.7$$

$$X = 0.224$$

$$Y = 0.32$$

$$N = \mathbf{10.3\ stages}$$

To get N_{actual}, efficiency is taken into consideration. An assumption of 60% is used:

$$N_{actual} = \frac{N}{\eta}$$

$$N_{actual} = 17.2 = \mathbf{18\ stages}$$

Next, the column diameter and height are found. To get the column diameter, the Sauders Brown equation is used:

Sauders Brown: $u_f = C\sqrt{\frac{\rho_L - \rho_V}{\rho_V}}$

$$Tray\ Capacity\ Constant,\ C = 0.11\frac{m}{s}$$

$$u_{des}(@80\%\ flooding) = 0.8u_f$$

Next, everything is plugged into the diameter relationship:

$$D = \sqrt{\frac{4\dot{V}}{\pi \phi u_{des}}}$$

$$\phi(active\ area\ fraction) = 0.8$$

$$D = 0.8\ m = \mathbf{2.6\ ft}$$

Next, to find the height of the column, the following equation is used:

$$H_{active} = (N_{trays} - 1)S$$

$$S\ (tray\ spacing) = 24\ in = 0.61\ m = \mathbf{2\ ft}$$

$$H_{active} = 10.4\ m = \mathbf{34\ ft}$$

Add an additional 3 m for tops and bottoms equipment

$$H = 13.4\ m = \mathbf{43\ ft}$$

Preliminary Utilities

Table 16: Preliminary Utility Summary

Utilities	cost	yearly Rate	Units		Units
Electricity cents/Kwh	19.9	8518320	Kwh/year	1695145.68	$/year
Low pressure steam $/kwh	0.021888	16000000	Kwh/year	350208	$/year
Waste water	2	80776000	gal/yr	161552	$/yr
Green Hydrogen $/lb	1.545454545	52000000	lb/yr	80363636.36	$/year
CoMO catalyst $/MT	53000	90	MT/yr	4770000	$/year
Cooling water $/gal	0.04664304	360000000	gal/yr	16791494.4	$/year
Fuel Fired $/kwh	0.02	180536000	Kwh/year	3610720	$/year

The various equipment in the process utilizes these utilities. For electricity, equipment such as pumps and compressors are the ones that both use and generate the overall electricity rate. Moreover, cooling water is used repeatedly throughout the process in various exchangers, condensers, etc. The use of cooling water is widespread and flexible, as it can cool a stream up to 25 °C. Low-pressure steam was also used to heat streams in heaters, exchangers, reboilers, and other components in columns. The same for fired heat, which is used for temperatures that steam cannot handle, such as reboilers in the de-butanization and fractionation of stable oil. The reaction of bio-oil in the hydrotreater yields immense amounts of wastewater with close to 99% water purity. But because of these impurities, it must be treated and sent to wastewater treatment. At a cost of $ 2 per 1,000 gallons and about 45,000 gallons/hr of wastewater produced in the process, a hefty cost will be added to the wastewater. The pricing of these utilities was obtained from Dr. C, the EIA, and Mr. Bahr as sources. All in all, the utilities listed in the chart above are preliminary and subject to change, as high-pressure steam can be generated rapidly during fast pyrolysis. As well as other utilities, they can be replaced with fuel gas from the de-butanizer tops and a possible source of electricity.

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